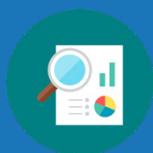


Unit 8 Overview: Proving Relationships and Theorems

Unit Narrative



Unit 8 provides students the opportunity to explore quadrilateral characteristics and properties. Students will gain experience in identifying the characteristics of specific quadrilaterals in terms of their relationship involving their sides, angles, and diagonals. Students will manipulate quadrilateral properties to prove theorems as a means to identify special parallelograms. In addition, they will use these properties to solve problems involving parallelograms, trapezoids, and kites. Other connections that are outlined in the unit build upon students' ability to solve for a variable, in order to determine values of side lengths and angle measures. This is an essential concept that students have been exposed to throughout prior mathematics courses.

In Section 1, students are focused on proving relationships and theorems about parallelograms. In Lesson 1, students define a parallelogram and identify the properties of a parallelogram using angles, sides, and diagonals. In Lesson 2, students use the properties discovered in Lesson 1 to determine if a quadrilateral is a parallelogram. Lesson 3 focuses students on using properties and theorems to prove if rectangles, rhombi, and squares are parallelograms. In Lesson 4, students will explore if a kite can be classified as a parallelogram, and use properties and theorems learned thus far in the unit to solve problems or prove certain characteristics or relationships in quadrilaterals.

Section 2 transitions students to prove relationships and theorems about trapezoids and solve real-world problems with quadrilaterals. Similar to the lessons in Section 1, in Lesson 5 students will prove relationships and theorems about trapezoids and use them to solve problems involving trapezoids. Lesson 6 is a culmination of the unit, where students will demonstrate their understanding of proving relationships and theorems for parallelograms and trapezoids to solve real-world problems. Through an Exploratory activity in Lesson 7, students discover the trapezoid midsegment theorem and apply their learning to solve mathematical and real-world problems.

Upon completion of Unit 8, students will be able to construct geometric proofs and solve problems involving the quadrilaterals covered within the unit. Students will build strong arguments by providing evidence such as side lengths, angle measures, and congruence properties and theorems in order to support their conclusions on classifying quadrilaterals.

Unit At-A-Glance

Section 1: Parallelograms	Benchmarks Addressed	Lesson 1	Proving Properties of Parallelograms
	MA.912.GR.1.4 MA.912.LT.4.10	Lesson 2	Using Properties of Parallelograms
		Lesson 3	Special Parallelograms
		Lesson 4	Kites and Parallelograms
Section 1: Trapezoids and other Quadrilaterals	Benchmarks Addressed	Lesson 5	Trapezoids
	MA.912.GR.1.4 MA.912.GR.1.5 MA.912.LT.4.10	Lesson 6	Solving Real-World Problems With Quadrilaterals
		Lesson 7	Trapezoid Midsegment Theorem



Differentiation

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Engagement <i>The “WHY” of learning</i>	In Lessons 1, 5, and 7, students are engaged in attention grabbing Notice and Wonder Routines. These Warm-Ups will foster engagement by eliciting the artistic thinking of students. Allow students the opportunity to think and share throughout the unit where they have seen parallelograms and/or trapezoids in the real-world, possibly in a sculpture they have seen, or in the fence they have in their yard.
Representation <i>The “WHAT” of learning</i>	Consider bringing in examples (or allowing students to bring in examples) of different rectangles, squares, rhombi, kites, and trapezoids. For example, during Lessons 1 and 4, different kites could be brought to class for students to be able to see how they are different from the other figures covered in the unit. Some students may have never seen a kite, or may not know what a kite is/does in the real-world.
Action & Expression <i>The “HOW” of learning</i>	- This unit will require students to deepen their understanding of different quadrilaterals. Students could create a mobile of things they currently know about quadrilaterals, and add to each item as they learn more. This could be accomplished by adjusting “The Boxes Go Mobile” to include information about each shape they explore during the unit. Students could create 2-D representations to cut out of each key word, and write the word on the shape. They then add to the shapes with new paper as they learn new words and features. Students should be challenged to continually keep their mobile balanced throughout the unit. - As a project, students could be asked to construct a kite after learning of its properties, and the kites could be hung around the room or in the hallways.

ELL Information

Create a visual aid (chart paper sized and interactive notebook sized) with a Venn diagram for quadrilaterals. Use several differently colored pencils and markers for each of the ovals, representing each type of quadrilateral with a different color. Additionally, create a chart (chart paper sized and interactive notebook sized) for keeping track of the different characteristics of quadrilaterals.

Provide as many other visual aids as possible to remind students of vocabulary, such as “opposite angle,” “opposite sides,” “adjacent sides,” “diagonals,” etc.

Create small groups and assign each group to create a poster with the name of a different type of quadrilateral, and illustrate everything they know about it; allow students to revise each poster as you progress through all the lessons on quadrilaterals.

Point out roots or prefixes that are cognates. For example, a “quadrilateral” has four sides, and “cuatro” means “four,” or “rectángulo” and “rectangle.” Sometimes, pronunciation is similar even if the spelling is not.

Allow students to work in partners or triads, when appropriate; carefully plan small groups so that ELL students are partnered with non-ELL students.

Provide frequent opportunities for students to turn-and-talk; pair up ELL students with non-ELL students.

Encourage students to make dual-language flashcards of the vocabulary words, properties, and theorems, including diagrams where appropriate.

All glossary terms are available in multiple languages, including English, Spanish, Haitian Creole, and American Sign Language, as support for ELL students. Consider pairing the following videos with strategies from “Additional Differentiated Support” below:

- Parallelogram
- Rectangle
- Rhombus
- Square
- Trapezoid

Additional Differentiated Support

Scaffolding Resources	On-Ramp
Quizlet offers an engaging and competitive activity where students have to identify the quadrilateral, or select what statement completes a proof of a parallelogram or trapezoid. The faster students can answer the question, the more points they get. This should be beyond identifying the type of quadrilateral, as the questions should primarily focus on proving relationships and theorems about parallelograms and trapezoids. Desmos offers an interactive and engaging game of “guess who” using quadrilaterals. Students are paired up randomly, and have to guess which quadrilateral their opponent picked. Students need to ask their opponent yes or no questions to help narrow down their list of quadrilaterals. This can be modified to where students asking questions about “does your quadrilateral have consecutive sides?” and the partner would answer, “yes” or “no.” This would again focus on the properties and relationships, as opposed to simply naming the types of quadrilaterals covered in the unit.	Use the videos and practice problems in the On-Ramp to Geometry personalized learning tool to review the following topics for this unit. Videos are available in English and Spanish. • Classifying Quadrilaterals • Classifying Other Two-Dimensional Shapes



Section Overview

Section 1: Parallelograms (Lessons 1-4)

Benchmarks Addressed	Learning Targets	
MA.912.GR.1.4 Prove relationships and theorems about parallelograms. Solve mathematical and real-world problems involving postulates, relationships and theorems of parallelograms. <i>Clarification 1: Postulates, relationships and theorems include opposite sides are congruent, consecutive angles are supplementary, opposite angles are congruent, the diagonals of a parallelogram bisect each other, and rectangles are parallelograms with congruent diagonals.</i> <i>Clarification 2: Instruction includes constructing two-column proofs, pictorial proofs, paragraph and narrative proofs, flow chart proofs or informal proofs.</i> <i>Clarification 3: Instruction focuses on helping a student choose a method they can use reliably.</i> MA.912.LT.4.10 Judge the validity of arguments and give counterexamples to disprove statements. <i>Clarification 1: Within the Geometry course, instruction focuses on the connection to proofs within the course.</i>	Lesson 1	- I can define a parallelogram. - I can prove opposite sides and opposite angles of a parallelogram are congruent. - I can prove consecutive angles of a parallelogram are supplementary. - I can prove diagonals of a parallelogram bisect each other.
	Lesson 2	I can determine if a quadrilateral is a parallelogram using properties of a parallelogram.
	Lesson 3	- I can prove rectangles are parallelograms. - I can use properties of parallelograms to identify special parallelograms.
	Lesson 4	I can use properties of a parallelogram, kites, and special parallelograms to solve mathematical and real-world problems.

Section 2: Trapezoids and Other Quadrilaterals (Lessons 5-7)

Benchmarks Addressed	Learning Targets	
MA.912.GR.1.4 Prove relationships and theorems about parallelograms. Solve mathematical and real-world problems involving postulates, relationships and theorems of parallelograms. <i>Clarification 1: Postulates, relationships and theorems include opposite sides are congruent, consecutive angles are supplementary, opposite angles are congruent, the diagonals of a parallelogram bisect each other, and rectangles are parallelograms with congruent diagonals.</i> <i>Clarification 2: Instruction includes constructing two-column proofs, pictorial proofs, paragraph and narrative proofs, flow chart proofs or informal proofs.</i> <i>Clarification 3: Instruction focuses on helping a student choose a method they can use reliably.</i> MA.912.GR.1.5 Prove relationships and theorems about trapezoids. Solve mathematical and real-world problems involving postulates, relationships and theorems of trapezoids. <i>Clarification 1: Postulates, relationships and theorems include the Trapezoid Midsegment Theorem and for isosceles trapezoids: base angles are congruent, opposite angles are supplementary and diagonals are congruent.</i> <i>Clarification 2: Instruction includes constructing two-column proofs, pictorial proofs, paragraph and narrative proofs, flow chart proofs or informal proofs.</i> <i>Clarification 3: Instruction focuses on helping a student choose a method they can use reliably.</i> MA.912.LT.4.10 Judge the validity of arguments and give counterexamples to disprove statements. <i>Clarification 1: Within the Geometry course, instruction focuses on the connection to proofs within the course.</i>	Lesson 5	- I can prove relationships and theorems about trapezoids. - I can use relationships and theorems about trapezoids.
	Lesson 6	- I can use properties of parallelograms to solve real-world problems. - I can use properties of trapezoids to solve real-world problems.
	Lesson 7	- I can prove the trapezoid midsegment theorem. - I can use properties of trapezoids to solve mathematical and real-world problems.



Vertical Alignment

Geometry Unit 8 Proving Relationships and Theorems

Prior Course(s)	Course Level
N/A	MA.912.GR.1.4 Prove relationships and theorems about parallelograms. Solve mathematical and real-world problems involving postulates, relationships, and theorems of parallelograms. MA.912.GR.1.5 Prove relationships and theorems about trapezoids. Solve mathematical and real-world problems involving postulates, relationships, and theorems of trapezoids.

8.1 Proving Properties of Parallelograms

Lesson Introduction

 Benchmarks	MA.912.GR.1.4 Prove relationships and theorems about parallelograms. Solve mathematical and real-world problems involving postulates, relationships, and theorems of parallelograms.		 Learning Targets	- I can define a parallelogram. - I can prove opposite sides and opposite angles of a parallelogram are congruent. - I can prove consecutive angles of a parallelogram are supplementary. - I can prove diagonals of a parallelogram bisect each other.
	MA.912.LT.4.10 Judge the validity of arguments, and give counterexamples to disprove statements.			
 Benchmark Coherence	Building On		Working Toward	
	N/A		N/A	
 Essential Questions	- What defines a figure as a parallelogram? - What are the properties of a parallelogram? - How do we prove that a given quadrilateral is a parallelogram?			
 Lesson Narrative	Students explore parallelograms by measuring sides, angles, and diagonals to make conjectures about the properties of parallelograms. Using definitions, postulates, and theorems, students will complete proofs involving parallelograms.			
 Component Summary	8.1.1 Warm-Up (~5 min.)	Review a career video of a Cartographer and respond to questions about maps. (SE p. 7)	8.1.3 Exploration (15-20 min.)	Write conjectures about the properties of parallelograms. (SE p. 8)
	8.1.2 Collaboration (5 min.)	Draw and classify a four-sided, enclosed shape. (SE p. 8)	8.1.4 Guided Instruction (10-15 min.)	Complete proofs about parallelograms. (SE p. 11)
	8.1.5 Wrap-Up (~5 min.)	Reflect on parallelograms by providing a fact that was known prior to the lesson, and a new fact that was learned. (SE p. 12)		
 Resources	Glossary		Materials	
	<u>Key Terms:</u> - Parallelogram <u>Additional Terms:</u> - Kite		- Patty Paper - Ruler - Protractor	
			<u>Additional Preparation</u> 8.1.1 Warm-Up contains a video to accompany the question prompts	

 Teacher Tips	Common Misconceptions - Students may assume that the measure of all angles of the shape will always be 90° for 8.1.3 Exploration, question 3. - Students may state a segment bisector cuts a segment into congruent parts. The definition of segment bisector states equal parts, not congruent. This is important to note for proper use later in writing proofs.	Things to Consider Consider adding the key information about parallelograms from this lesson to a proof resource booklet, a foldable, or flipbook, if one was started prior to this unit. If not, consider starting one and naming it “parallelograms.”
	Literacy and Language Routines Think-Pair-Share Consider utilizing a Think-Pair-Share routine for 8.1.4 Guided Instruction, questions 2 and 3. This will allow students the opportunity to independently approach each proof, and discuss their thinking before teacher facilitation.	Mathematical Thinking and Reasoning (MTR) Standard MA.K12.MTR.2.1: Multiple Representations Students are positioned in 8.1.3 Exploration, question 5 to develop illustrations highlighting the different properties of parallelograms. This is an opportunity for students to demonstrate understanding of the content through drawing and naming parallelograms, and writing a mathematical statement for the picture they drew.
	 Differentiate Instruction Provide an entry point into the proofs in 8.1.4 Guided Instruction by providing a list of all the statements and reasons on cards. Students can use the cards to find opportunities to gain entry into the lesson.	
Lesson Notes:		

8.1.1 Warm-Up: Cartographer

Component Overview: Play the career profile video on the Cartographer for the entire class. After students watch the video (4:12 minutes), prompt them to reflect on the profession by answering the two questions independently.



8.1.1 Warm-Up: Cartographer

Dave Imus is a cartographer whose maps have won many awards. Dave sees the world through understanding places. Dave's maps are considered art.

1. Have you ever used a paper map?

Answers will vary. No, I use my GPS to find my way to a new place.

2. Since you can't put everything on a map, what three places do you think should be on the map of Florida?

Answers will vary. I think Disney World, Pensacola, and Daytona Speedway.



Consider bringing in different paper maps of Florida or different states, so that students can see and experience a map in person.

8.1.2 Collaboration: Parallel Lines and Four-Sided Figures

Component Overview: Using patty paper, students will create parallelograms from two set of parallel lines. Students will classify the type of parallelogram they created, and justify their classification.



8.1.2 Collaboration: Parallel Lines and Four-Sided Figures

1. On a piece of patty paper, draw one set of parallel lines. Overlap your patty paper with a partner's parallel lines to create an enclosed figure. Each partner should trace the enclosed shape onto a new piece of patty paper.
 - a. Classify the enclosed shape.
parallelogram
 - b. Justify your classification.
A parallelogram is a four-sided shape with opposite sides that are parallel.
 - c. Ask two classmates how they classified the figure they created. Record their classification and write your summary of their justification. *You are the only person who should write in the first two columns of the table.* Have the person initial next to your summary stating that your summary was correct.
Answers will vary.

Classification	My Summary of Their Justification	Initials



If needed, refer students back, very briefly, to prior lessons on creating parallel lines. Focus on the definition of parallel lines, and angle relationships formed by parallel lines and a transversal.



Students may draw these lines very close to each other. Remind students to leave at least 10 cm between the lines and continue the lines to the very edge of the paper.

8.1.3 Exploration: Properties of Parallelograms

Component Overview: Using the parallelograms created during the Exploration, students will measure segment lengths and angles to determine the properties of parallelograms. The students will then represent the properties in words, pictorially, and using mathematical statements.



8.1.3 Exploration: Properties of Parallelograms



A quadrilateral containing two pairs of parallel sides is a **parallelogram**. A member of the following shape classes: polygons, quadrilaterals, trapezoids. Sub-classes include rectangles, rhombi, and squares.

1. Using a ruler, measure the pairs of opposite sides in each parallelogram you and your partner created.
 - a. What do you notice?
They are the same length.
 - b. Make a conjecture about the measures of opposite sides of a parallelogram.
Opposite sides of a parallelogram are congruent.
2. Using a ruler, draw the diagonals of one of the parallelograms.
 - a. Determine the lengths of each diagonal.
Answers will vary.
 - b. Mark the point where the two diagonals intersect.
Answers will vary.
 - c. What do you notice about the point of intersection and its relationship to each diagonal?
The point of intersection is the midpoint of both diagonals.
 - d. Repeat the same steps to draw and measure the diagonals in your second parallelogram.
Answers will vary.
 - e. Make a conjecture about the diagonals of a parallelogram.
Each diagonal of a parallelogram bisects the other diagonal.



Students may believe that all the angle measures will always be 90° . Consider asking students to draw the shape differently to see if the angles are still right angles.

8.1.3 Exploration: Properties of Parallelograms (continued)

3. Using a protractor, measure the pairs of opposite interior angles of your parallelograms.
 - a. What do you notice?
The opposite interior angles have the same measure.
 - b. Make a conjecture about the measures of opposite angles of a parallelogram.
The opposite interior angles of a parallelogram are congruent.
 - c. Make a conjecture about the measures of consecutive angles of a parallelogram.
Consecutive angles of a parallelogram are supplementary angles.
4. Based on your conjectures, select all of the statements about a parallelogram that are true.
 - ☐ A parallelogram has only one set of opposite parallel sides.
 - ☐ *A parallelogram has two sets of opposite parallel sides.*
 - ☐ A parallelogram has only one set of opposite congruent sides.
 - ☐ *A parallelogram has two sets of opposite congruent sides.*
 - ☐ Consecutive angles of a parallelogram are congruent.
 - ☐ *Consecutive angles of a parallelogram are supplementary.*
 - ☐ *Opposite angles of a parallelogram are congruent.*
 - ☐ Opposite angles of a parallelogram are supplementary.
 - ☐ *Diagonals of a parallelogram bisect each other.*

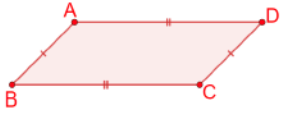
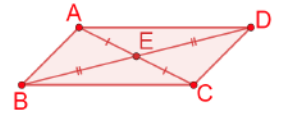


True or False

Consider utilizing this routine when question 4 by displaying one statement at a time and having students indicate their answers using thumbs up/thumbs down or holding up cards. As students share why they selected their answer, prompt them to justify their response, whether right or wrong, with correct mathematical vocabulary. Listen for words and phrases like “transversal,” “distance between lines”, “equal length,” etc.

8.1.3 Exploration: Properties of Parallelograms (continued)

5. For each true statement in the previous problem, draw and name a parallelogram. Using the information in the statement, mark the diagram and write a mathematical statement. The first row has been completed.

Statement	Drawing	Given
A parallelogram has two sets of opposite parallel sides.		$\overline{AD} \parallel \overline{BC}$ $\overline{AB} \parallel \overline{DC}$
A parallelogram has two sets of opposite congruent sides.		$\overline{AD} \cong \overline{BC}$, $\overline{AB} \cong \overline{DC}$
Consecutive angles of a parallelogram are supplementary.		$\angle ABC$ and $\angle BCD$ are supplementary, $\angle BCD$ and $\angle CDA$ are supplementary, $\angle CDA$ and $\angle DAB$ are supplementary, $\angle DAB$ and $\angle ABC$ are supplementary
Opposite angles of a parallelogram are congruent.		$\angle ABC \cong \angle CDA$, $\angle BCD \cong \angle DAB$
Diagonals of a parallelogram bisect each other.		$AE = CE$, $BE = DE$



MA.K12.MTR.2.1: Multiple Representations

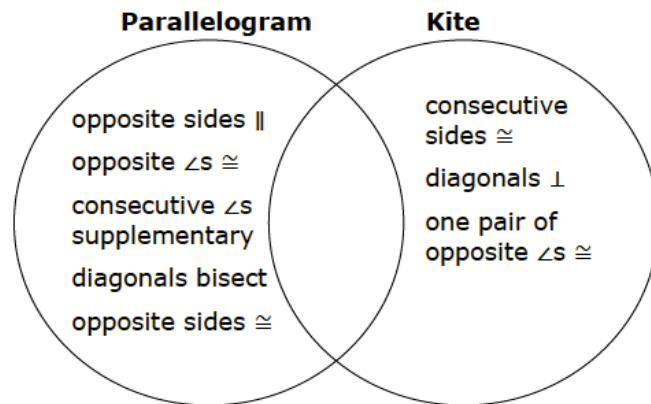
The design of this activity allows students to represent the properties of parallelograms in multiple ways. Students begin identifying each statement by the words. Using this, students are invited to develop a pictorial and mathematical representation of each property.



Students may write the mathematical statement using “congruence,” instead of “equality.” Have students recall the definition of segment bisectors from Unit 1 to determine which to use.

8.1.3 Exploration: Properties of Parallelograms (continued)

6. The Venn diagram shows the properties of a parallelogram and another quadrilateral called a kite. Discuss with your partner why a kite is not considered a parallelogram.



Students have not been exposed to the vocabulary word “kite.” This word is defined in Lesson 4.

Students are not required to write a summary of their discussion. While monitoring student conversation, listen for students to connect that a figure can’t be both a parallelogram and a kite, but could be one or the other.



Theorems about Parallelograms

- Opposite sides of a parallelogram are congruent.
- Opposite angles of a parallelogram are congruent.
- Consecutive interior angles of a parallelogram are supplementary.
- The diagonals of a parallelogram bisect each other.

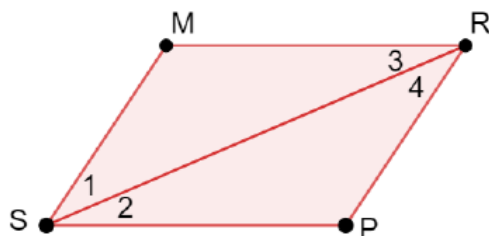
8.1.4 Guided Instruction: Proofs With Parallelograms

Component Overview: Guide students through proving the theorems related to the properties of parallelograms. The intent is to complete these proofs whole-group, allowing students to take the lead in filling in the statements and reasons.



8.1.4 Guided Instruction: Proofs with Parallelograms

1. $MRPS$ is shown.



Given: $MRPS$ is a parallelogram

Prove: $\overline{MR} \cong \overline{PS}$ and $\overline{MS} \cong \overline{PR}$

Complete the following two-column proof.

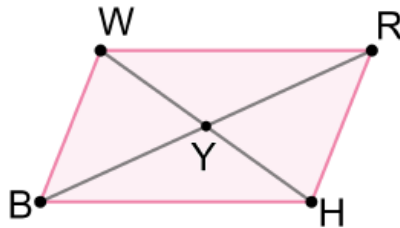
Statement	Reason
1. $MRPS$ is a parallelogram	1. Given
2. $\overline{MR} \parallel \overline{SP}, \overline{MS} \parallel \overline{RP}$	2. Definition of a parallelogram
3. $\angle 1 \cong \angle 4, \angle 2 \cong \angle 3$	3. If two parallel lines are intersected by a transversal, then alternate interior angles are congruent.
4. $\overline{SR} \cong \overline{SR}$	4. Reflexive property of congruence
5. $\triangle MSR \cong \triangle PRS$	5. ASA congruence theorem
6. $\overline{MR} \cong \overline{PS}$ and $\overline{MS} \cong \overline{PR}$	6. CPCTC



As an additional scaffold, consider providing the possible statements and reasons on note cards for students who still struggle with writing proofs to choose from.

8.1.4 Guided Instruction: Proofs With Parallelograms (continued)

2. $WRHB$ with diagonals $\overline{WH}$ and $\overline{BR}$ intersecting at Y is shown.



Given: $WRHB$ is a parallelogram with diagonals $\overline{WH}$ and $\overline{BR}$ intersecting at Y .

Prove: $\overline{WH}$ and $\overline{BR}$ bisect each other.

Complete the following two-column proof.

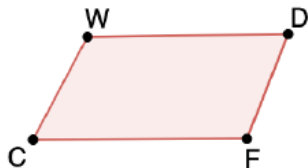
Statement	Reason
1. $WRHB$ is a parallelogram with diagonals $\overline{WH}$ and $\overline{BR}$ intersecting at Y .	1. Given
2. $\overline{WR} \parallel \overline{BH}$	2. Definition of a parallelogram
3. $\overline{RH} \cong \overline{BW}$; $\overline{WR} \cong \overline{HB}$	3. Opposite sides of a parallelogram are congruent.
4. $\angle WRB \cong \angle HBR$; $\angle BHW \cong \angle RWH$	4. If two parallel lines are intersected by a transversal, then alternate interior angles are congruent.
5. $\triangle BYH \cong \triangle RYW$	5. ASA congruence theorem
6. $\overline{BY} \cong \overline{RY}$ and $\overline{WY} \cong \overline{YH}$	6. CPCTC
7. $BY = YR$ and $WY = YH$	7. Definition of congruence
8. $\overline{WH}$ and $\overline{BR}$ bisect each other.	8. Definition of a segment bisector



Students may include both sets of parallel lines in the proof. Facilitate a discussion on why it is not necessary to include $\overline{WB} \parallel \overline{RH}$. Ask probing questions like, "What angles are congruent by stating these two lines are parallel? Are these angles needed to prove $\triangle BYH \cong \triangle RYW$?"

8.1.4 Guided Instruction: Proofs With Parallelograms (continued)

3. Complete the following flowchart proof. Statements go in the boxes and reasons go below the boxes.

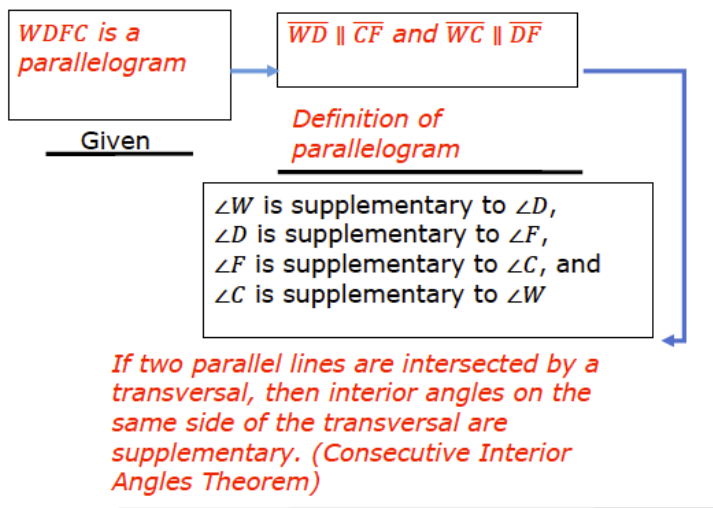


Given: $WDFC$ is a parallelogram.

Prove: $\angle W$ is supplementary to $\angle D$,
 $\angle D$ is supplementary to $\angle F$,
 $\angle F$ is supplementary to $\angle C$, and
 $\angle C$ is supplementary to $\angle W$



Consider allowing them to complete question 3 as a Think-Pair-Share.



8.1.5 Wrap-Up: Cool Down

Component Overview: Students reflect on what they knew already about parallelograms, if anything, and identify something new that they learned.



8.1.5 Wrap-Up: Cool Down

1. Write down one fact you knew about parallelograms before the lesson.

Answers will vary.

2. Write down one fact you learned about parallelograms during the lesson.

Answers will vary.

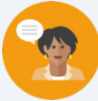


If there is extra time, ask students to form their answers to question 2 into assessment questions. Ask students to write these questions on note cards, and use them in a future review lesson.

8.2 Using Properties of Parallelograms

Lesson Introduction

 Benchmarks	MA.912.GR.1.4 Prove relationships and theorems about parallelograms. Solve mathematical and real-world problems involving postulates, relationships, and theorems of parallelograms.		 Learning Target	I can determine if a quadrilateral is a parallelogram using properties of a parallelogram.
	MA.912.LT.4.10 Judge the validity of arguments, and give counterexamples to disprove statements.			
 Benchmark Coherence	Building On		Working Toward	
	N/A		N/A	
 Essential Questions	- What qualities must a figure have to be a parallelogram? - How do we use properties of a parallelogram to prove a quadrilateral is a parallelogram?			
 Lesson Narrative	In this lesson, students will use the properties of parallelograms to determine if a quadrilateral is a parallelogram. Given information is provided, including in real-world scenarios, for students to examine and decide if it is enough to state the quadrilateral must be a parallelogram.			
 Component Summary	8.2.1 Warm-Up (~ 5 min.)	Complete a proof to prove if a figure is a parallelogram. (SE p. 13)	8.2.3 Guided Practice (10-15 min.)	Determine if the given information is enough to prove a figure is a parallelogram. (SE p. 16)
	8.2.2 Collaboration (10-15 min.)	Determine if quadrilaterals are parallelograms. (SE p. 14)	8.2.4 Try It (10 min.)	Use Algebra to determine if a figure is a parallelogram. (SE p. 17)
	8.2.5 Wrap-Up (~ 5 min.)	Choose properties that determine if a quadrilateral is a parallelogram. (SE p. 18)		
 Resources	Glossary		Materials	Additional Preparation
	<u>Key Terms:</u> N/A <u>Additional Terms:</u> - Quadrilateral - Parallelogram		(Optional) Rubber bands (Optional) Geo Boards (Optional) Sticks (Optional) String (Optional) Rulers (Optional) Protractors	N/A

 Teacher Tips	Common Misconceptions • Students may perform incorrect calculations when substituting values for x and y in 8.2.4 Try It. • Students may set the wrong expressions equal to each other in 8.2.2 Collaboration and 8.2.4 Try It.	Things to Consider • Consider using 8.2.1 Warm-Up as a formative assessment of the content understood and retained from the prior lesson.
	Literacy and Language Routines Decide and Defend For 8.2.2 Collaboration, consider utilizing a Decide and Defend routine with student partners. To utilize this routine, have students respond to each figure as to if it is a parallelogram, and have them defend their decision. Have partners (or pairs of partners) share their decisions, along with their defense statements (justifications). Allow time for students to go back and revise their answers after listening to their peers' defense.	Mathematical Thinking and Reasoning (MTR) Standard MA.K12.MTR.6.1: Reasonableness 8.2.2 Collaboration provides students the opportunity to draw conclusions about a figure and assess the reasonableness of their answer by responding to “How do you know?” through justifying.
	 Differentiate Instruction	
Lesson Notes:		

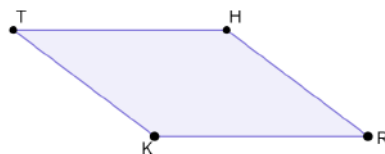
8.2.1 Warm-Up: Bell Work

Component Overview: Students will complete the proof to prove the quadrilateral with the given information is a parallelogram. Use the data gathered from this activity as a formative assessment of students' understanding of the content presented in Lesson 1.



8.2.1 Warm-Up: Bell Work

Given: $\angle RKT \cong \angle RHT$ and
 $\angle KTH \cong \angle KRH$
 Prove: $THRK$ is a
 parallelogram.



Statement	Reason
1. $\angle RKT \cong \angle RHT$ and $\angle KTH \cong \angle KRH$	1. Given
2. $m\angle RKT = m\angle RHT$ and $m\angle KTH = m\angle KRH$	2. Definition of Congruence
3. $m\angle RKT + m\angle RHT + m\angle KTH + m\angle KRH = 360$	3. The sum of the angles in a quadrilateral is 360° .
4. $m\angle RKT + m\angle RKT + m\angle KTH + m\angle KTH = 360^\circ$ $m\angle RHT + m\angle RHT + m\angle KRH + m\angle KRH = 360^\circ$	4. <i>Substitution property of equality</i>
5. $2m\angle RKT + 2m\angle KTH = 360^\circ$ $2m\angle RHT + 2m\angle KRH = 360^\circ$	5. Addition
6. $m\angle RKT + m\angle KTH = 180^\circ$ $m\angle RHT + m\angle KRH = 180^\circ$	6. <i>Division Property of Equality</i>
7. $\angle RKT$ is supplementary to $\angle KTH$. $\angle RHT$ is supplementary to $\angle KRH$.	7. <i>Definition of Supplementary</i>
8. $\overline{TH} \parallel \overline{KR}$ and $\overline{TK} \parallel \overline{HR}$	8. If two lines are intersected by a transversal so that consecutive interior angles are supplementary, then the lines are parallel.
9. $THRK$ is a parallelogram.	9. <i>Definition of a Parallelogram</i>



Use students' understanding to inform your instruction and to determine where during this and further lessons. Make a list of misconceptions and determine which should be addressed in the whole group, and which should be addressed in a small group with those who are struggling to be proficient.

8.2.2 Collaboration: Determining Parallelograms

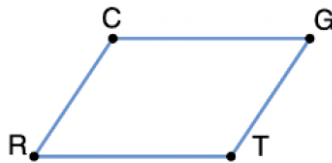
Component Overview: Students use given information to make conclusions about a series of four-sided figures. Students will justify their conclusions. Additionally, students will identify the value of two variables that will make a figure a parallelogram. Students conclude the Collaboration by summarizing the qualities a figure must have to be a parallelogram.



8.2.2 Collaboration: Determining Parallelograms

Work with your partner to determine if the figure given is a parallelogram. Finish the statements that follow.

1. $CGTR$ is shown. In the figure, $m\angle C = 113^\circ$, $m\angle G = 67^\circ$ and $m\angle T = 103^\circ$.



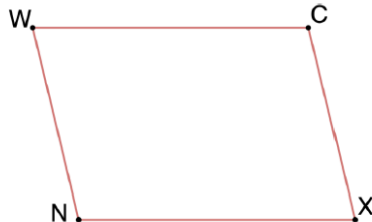
- a. I can conclude ...

that quadrilateral $CGTR$ is not a parallelogram.

- b. I know this because ...

$\angle C$ and $\angle T$ are opposite angles, but they are not congruent.

2. $WCXN$ is shown with sides $\overline{WC}$ and $\overline{NX}$ each measuring 15 cm and sides $\overline{WN}$ and $\overline{CX}$ each measuring 9 cm.



- a. I can conclude ...

that quadrilateral $WCXN$ is a parallelogram.

- b. I know this because ...

there are two pairs of opposite sides of equal length.



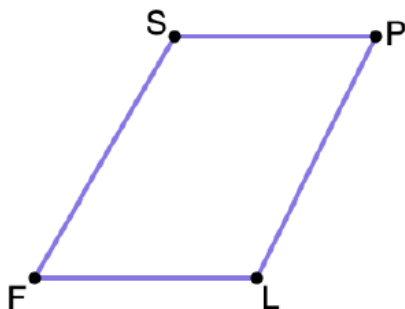
What could we change about the given information to make this figure a parallelogram?



Provide students materials to create their own quadrilateral that would fulfill the description. Supplies could include geoboards, rubber bands, string, sticks, and a ruler.

8.2.2 Collaboration: Determining Parallelograms (continued)

3. In $SPLF$, $m\angle S = 120^\circ$, and $m\angle F = (20x + 47)^\circ$. If $x = \frac{1}{2}$, determine whether the figure is a parallelogram.



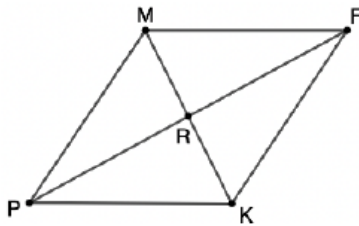
- a. I can conclude ...
 $SPLF$ is not a parallelogram.
- b. I know this because ...
 $\angle S$ and $\angle F$ are consecutive angles but are not supplementary.



After students have determined that the quadrilateral is not a parallelogram because the angles would not be supplementary, ask, "If the two angles were supplementary, would that be enough information to prove the quadrilateral is a parallelogram? (Answer: No; if $x = 0.65$, the two angles would be supplementary, but that only proves that one set of sides is parallel.)"

8.2.2 Collaboration: Determining Parallelograms (continued)

4. $MFKP$ with diagonals $\overline{MK}$ and $\overline{PF}$ intersecting at point R is shown.



Given that $MR = 4x + 7$, $KR = 3y - 2$, $FR = 4y - 3$, and $PR = 7x - 1$, find the values of x and y that make $MFKP$ a parallelogram.

$$x = \underline{\quad 6 \quad} \quad y = \underline{\quad 11 \quad}$$



Students may incorrectly set $\overline{MR} = \overline{PR}$, and $\overline{KR} = \overline{FR}$, because they share the same variable. After students create the correct equations, provide an opportunity for students to recall the methods for solving a system of equations by asking probing questions.

5. When determining if a figure is a parallelogram, what qualities must it have? List them below.

Opposite sides are parallel.

Opposite sides are congruent.

Opposite angles are congruent.

Consecutive angles are supplementary.

The diagonals of a parallelogram bisect each other.

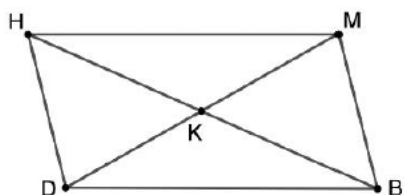
8.2.3 Guided Practice: Is It a Parallelogram?

Component Overview: Students justify whether given information is enough to determine if a quadrilateral is a parallelogram.



8.2.3 Guided Practice: Is It a Parallelogram?

- Use the figure given to fill in the chart below.



If you knew ...	...would HMBD be a parallelogram?	Why or why not?
$\angle DHK \cong \angle MBK$ and $\angle BDK \cong \angle HMK$	☐ yes ☐ no	Parallel lines have congruent alternate interior angles. Two sets of parallel lines define a parallelogram.
$\overline{HM} \cong \overline{DB}$	☐ yes ☐ no	Both sets of opposite sides must be congruent, not just one set.
K is the midpoint of $\overline{DM}$ and $\overline{HB}$	☐ yes ☐ no	The midpoint is the point where two segments will bisect each other. Parallelograms have diagonals that bisect each other.
$m\angle DHK + m\angle BDK = 180^\circ$	☐ yes ☐ no	These angles are not an angle pair formed by parallel lines.



Use observational data from the 8.2.2 Collaboration to determine how much guidance is needed to address key ideas in this Guided Practice component.

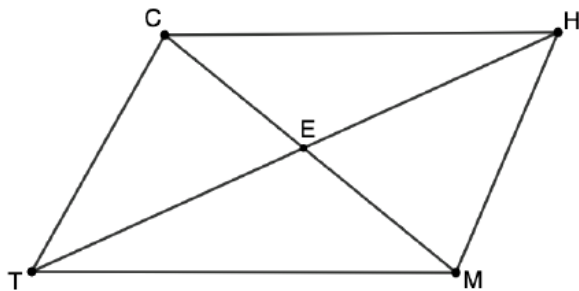


Co-Craft Questions and Problems

Consider utilizing this routine for each set of given information that had the response of "no." Students will create assessment questions, along with the correct response, determining what additional information is needed with the given information to determine the figure is a parallelogram.

8.2.3 Guided Practice: Is It a Parallelogram? (continued)

2. Rocio recently inherited a large field in the shape of a quadrilateral from her Uncle Asher. $CHMT$ with diagonals $\overline{CM}$ and $\overline{HT}$ intersecting at point E is shown. She wants to partition the field into four triangular sections to make gardens. While she was out measuring for her supply order, she noticed that the field was shaped like a parallelogram. Her Uncle Asher disagrees. Rocio created the diagram below with the following measurements: $m\angle CHT = 23^\circ$, $m\angle CET = 72^\circ$, $m\angle CMT = 49^\circ$, and $m\angle HTM = 23^\circ$.



- a. Who is correct?

Uncle Asher

- b. How do you know?

Only one set of lines can be proven parallel from the given information. Therefore the shape is not necessarily a parallelogram.



Consider challenging students to draw examples of quadrilaterals that have the given properties. Using student drawings, guide the discussion on why the quadrilateral does not necessarily have to be a parallelogram, but could also be a non-parallelogram trapezoid.

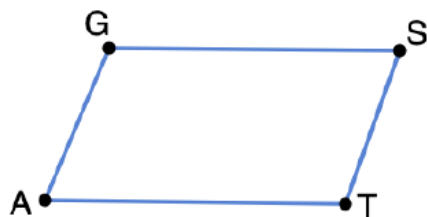
8.2.4 Try It: Parallelograms and Algebra

Component Overview: Students determine whether quadrilaterals are parallelograms using algebra to find unknown values.



8.2.4 Try It: Parallelograms and Algebra

1. Quadrilateral $GSTA$ is shown where $GS = 3x - 1$, $ST = 2y - 3$, $AT = 11$, and $GA = 8$.



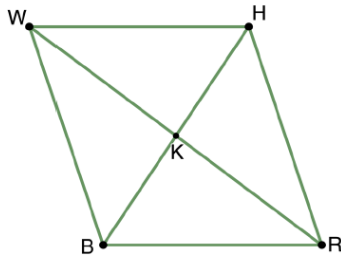
- a. If $x = 4$ and $y = 6$, determine whether $GSTA$ is a parallelogram.
No
- b. If $GSTA$ is not a parallelogram, what value(s) for x and y would make it a parallelogram?
 $x = 4$ and $y = 5.5$



Students may write the equation incorrectly by setting the wrong expressions equal to each other.

8.2.4 Try It: Parallelograms and Algebra (continued)

2. $WHRB$ is shown, where diagonals $\overline{WR}$ and $\overline{HB}$ intersect, such that $WK = -y + 4$, $RK = \frac{1}{3}x + 5$, $HK = 5x - 6$, and $BK = 11 + y$.

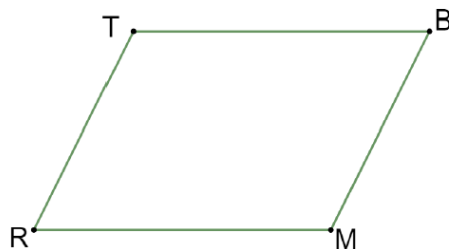


- a. If $x = 3$ and $y = -2$, determine whether $WHRB$ is a parallelogram.
Yes, it is.
- b. If $WHRB$ is not a parallelogram, what value(s) for x and y would make it a parallelogram?
No answer expected as the answer to part a is yes.



Consider allowing students to check their answers for the Try It with a partner.

3. Tariq and Agnes were having a discussion about the different properties of parallelograms. Together they drew and labeled $TBMR$, where $m\angle RTB = (5x - 2)^\circ$, $m\angle TBM = y^\circ$, and $m\angle BMR = 118^\circ$. What are the values of x and y for Tariq and Agnes' figure to be a parallelogram?



$x = 24$ and $y = 62$



If determined $y = 118$ and/or $x = 12.8$, then they used the wrong angle pair relationships, and set the wrong expressions equal to each other.

8.2.5 Wrap-Up: Cool Down

Component Overview: Students complete a checklist of properties that determine if a figure is a quadrilateral, a parallelogram, or both. Use the data collected in this activity to determine which students need remediation on the properties of parallelograms.



8.2.5 Wrap-Up: Cool Down

1. Check the property that is always true about the polygon.

Properties of Figure	Types of Polygons	
	Quadrilateral	Parallelogram
Only one set of parallel sides	☐	☐
Two sets of parallel sides	☐	☒
Two sets of sides of equal length	☐	☒
Consecutive angles are supplementary.	☐	☒
Diagonals are congruent.	☐	☐
The sum of all angles is 360° .	☒	☒
Diagonals bisect each other.	☐	☒
Opposite angles are congruent.	☐	☒



To provide an alternative entry point, have students draw as many different shapes as they can for each property.

8.3 Special Parallelograms

Lesson Introduction

 Benchmarks	MA.912.GR.1.4 Prove relationships and theorems about parallelograms. Solve mathematical and real-world problems involving postulates, relationships, and theorems of parallelograms.		 Learning Targets	- I can prove rectangles are parallelograms. - I can use properties of parallelograms to identify special parallelograms.
	MA.912.LT.4.10 Judge the validity of arguments and give counterexamples to disprove statements.			
 Benchmark Coherence	Building On		Working Toward	
	N/A		N/A	
 Essential Questions	- What properties determine that a parallelogram is a rectangle? - What properties determine that a parallelogram is a rhombus? - What properties determine that a parallelogram is a square?			
 Lesson Narrative	This lesson builds on the properties of parallelograms for students to explore the differences between rectangles, rhombi, and squares.			
 Component Summary	8.3.1 Warm-Up (~5 min.)	Compare an estimate to a calculated circumference. (SE p. 19)	8.3.4 Guided Instruction (10-15 min.)	Determine the properties of a square and a rhombus. (SE p. 22)
	8.3.2 Collaboration (10-15 min.)	Determine the properties of a rectangle. (SE p. 20)	8.3.5 Bring It Together (5 min.)	Explain how to determine if a parallelogram is a rectangle, square, or rhombus. (SE p. 24)
	8.3.3 Exploration (5-10 min.)	Compare properties of parallelograms and rectangles. (SE p. 21)	8.3.6 Wrap-Up (~5 min.)	Reflect on the interest and understanding of the lesson. (SE p. 25)
 Resources	Glossary		Materials	
	<u>Key Terms:</u> - Rectangle - Square - Rhombus <u>Additional Glossary:</u> N/A		- Rulers (Optional) String or Shoelace (Optional) Chart paper	
	<u>Additional Preparation</u> Make handouts of a blank chart that can be filled in to categorize polygons.			

 Teacher Tips	Common Misconceptions	Things to Consider
	- Students may confuse rectangles and squares. Drawing pictures with examples and non-examples can help students visualize the differences and deepening their understanding. - Students may place statements in more than one area of the Venn diagram in 8.3.4 Guided Instruction.	Consider having students create a foldable or use chart paper to display properties of each type of parallelogram discussed in the lesson, along with pictorial representations that are examples and non-examples.
	Literacy and Language Routines	Mathematical Thinking and Reasoning (MTR) Standard
 Differentiate Instruction	Stronger and Clearer Each Time During 8.3.5 Bring It Together, consider using the “Stronger and Clearer Each Time” routine to provide a structured and interactive opportunity for students to revise and refine both their ideas (verbal and written). This routine provides a purpose for student conversation, as well as fortifies the output. Throughout this process, students should be pressed for details, and encouraged to press each other for details.	MA.K12.MTR.4.1: Discussion and Reflection In 8.3.2 Collaboration and 8.3.3 Exploration, students will have the opportunity to discuss and reflect as they determine the attributes of rectangles and discover that rectangles are a type of parallelogram. Students will make conjectures about the lengths of diagonals in a rectangle, and about what happens at the intersection of diagonals in a rectangle, and then apply their understanding on rectangle $RPQD$ in question 2 during the 8.3.3 Exploration.
	This lesson is focused around developing clear definitions for different types of parallelograms. There are multiple opportunities throughout the lesson for kinesthetic learners to have an entry point, for example, when measuring segment lengths in 8.3.4 Guided Instruction. Consider allowing students voice and choice in how they respond to the questions in 8.3.5 Bring It Together. This could include students explaining to a partner, drawing diagrams with labels, recording an audio message, or writing an explanation.	
Lesson Notes:		

8.3.1 Warm-Up: Estimation

Component Overview: Students estimate and then use their understanding of circumference and diameter to determine the number of cereal pieces it would take to cover the edge of their bowl.



8.3.1 Warm-Up: Estimation

1. A picture of a bowl with a cereal square is shown.



- a. Estimate the number of cereal squares that will go around the inner circumference of the bowl.

Answers will vary.

40

2. A picture of the bowl with a diameter cereal squares is shown.



- a. Use the diameter of cereal squares to find the circumference.

43 or 44 cereal squares



This activity could be easily replicated in the classroom with paper bowls or plates, and any small box of cereal.

8.3.1 Warm-Up: Estimation (continued)

- b. Compare your estimate to the circumference found using the diameter of cereal squares.

Answers will vary. The diameter of the circle was 14 cereal squares. When calculating the circumference using it, the total was 43.982 which is just a little more than the estimation of the diameter of the cereal squares that could go around the bowl.

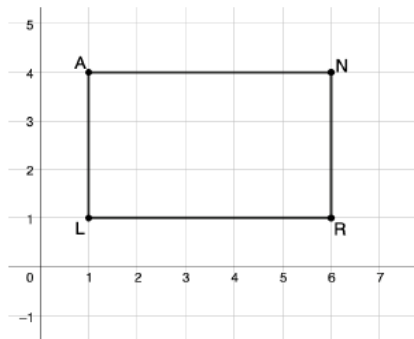
8.3.2 Collaboration: What Is a Rectangle?

Component Overview: Students will measure the lengths of various segments. Using these measurements, students will write a conjecture about the properties of rectangles. Students will need a partner for this component.



8.3.2 Collaboration: What is a Rectangle?

1. *ANRL* is a rectangle shown on the coordinate grid.



- a. List everything you know about this type of quadrilateral.
Answers will vary.
Two sets of parallel lines, opposite sides are congruent, and the sides intersect at right angles.
- b. Compare your list with your partner's list. Add anything you may have missed.
Answers will vary.
- c. With your partner, list two truths and one lie about this type of quadrilateral in the table below. Share your list with another pair. See if they can spot the lie.

Answers will vary.

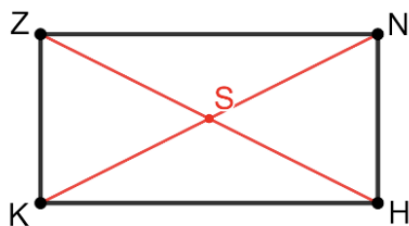
1. A rectangle has all sides congruent.
2. A rectangle has congruent angles.
3. A rectangle has two sets of congruent sides.



The rectangle shown is also a quadrilateral. Facilitate discussion on why a rectangle is a sub-class of a quadrilateral. Consider starting a chart that shows the classifications of polygons that is built upon through this lesson and future lessons.

8.3.2 Collaboration: What Is a Rectangle? (continued)

2. $ZNHK$ is shown.



- Using a ruler, draw $\overline{ZH}$ and $\overline{KN}$.
- These segments are called *diagonals*.
- Measure the length of these segments and write them in the table below. Include units.
Answers will vary depending on unit used, but they should be congruent.

Segment	Length
$\overline{ZH}$	7.8 cm
$\overline{KN}$	7.8 cm

- Make a conjecture about the lengths of the diagonals of a rectangle.
The lengths of the diagonals of a rectangle are equal.
- Label the intersection point of the diagonals S .
Shown on the figure.
- Measure the following segments and fill in their length in the table below. Include units.
Answers will vary depending on unit used, but they should be congruent.

Segment	Length
$\overline{ZS}$	3.9 cm
$\overline{HS}$	3.9 cm
$\overline{KS}$	3.9 cm
$\overline{NS}$	3.9 cm



Consider challenging some students to complete this task with only a piece of string or a shoelace. Students could tie knots in the string and measure the number of “sections” or units that they count for the measurement. The intent here is not to have the same measurement as someone else, but to realize that each segment has the same length, regardless of the unit of measure.



For ELL students, consider offering a word bank for question 2b, to allow for an entry point.

8.3.2 Collaboration: What Is a Rectangle? (continued)

- g. What do you notice?

Answers will vary.

All the lengths from the point of intersection on the diagonals are the same.

- h. Make a conjecture about what happens to the diagonals of a rectangle when they intersect.

The diagonals of a rectangle bisect each other.



While monitoring students' work, listen for appropriate mathematical vocabulary, like "point of intersection" and "bisect/bisector."

8.3.3 Exploration: Comparing Parallelograms and Rectangles

Component Overview: Students discover that a rectangle is a type of parallelogram and complete statements for a given rectangle.



8.3.3 Exploration: Comparing Parallelograms and Rectangles

1. Work with your partner to fill in the table. If the statement describes a property always true of the shape, mark the box.

Statement	Parallelogram	Rectangle
opposite sides congruent	✓	✓
opposite sides parallel	✓	✓
opposite angles congruent	✓	✓
consecutive angles supplementary	✓	✓
diagonals bisect each other	✓	✓
diagonals are congruent		✓
all angles are congruent		✓

Based on the chart, discuss with your partner whether a rectangle could be classified as a parallelogram.

Answers will vary.

Yes, a rectangle can be classified as a parallelogram because all the properties of a parallelogram are also properties of a rectangle.



A **rectangle** is a quadrilateral that contains four right angles. Rectangles may be oblong or square. It's a member of the following shape classes: polygons, quadrilaterals, trapezoids, parallelograms. Squares form a sub-class.



This Exploration is scaffolded to allow students to engage with their partner, rather than being guided through by the teacher.

Observe and listen as students work with their partners. Make note of students' ideas that you can ask them to share before moving to the next component.

8.3.3 Exploration: Comparing Parallelograms and Rectangles (continued)

2. $RPQD$ is a rectangle.



Complete the statements.

It is given that $RPQD$ is a rectangle. By the definition of

a rectangle $\angle R, \angle P, \angle Q$, and $\angle D$ are all

- ☐ right angles,
- ☐ acute angles,
- ☐ obtuse angles,

which means that $m\angle R = m\angle P = m\angle Q = m\angle D =$

- ☐ 90° .
- ☐ 180° .

Since $m\angle P + m\angle Q = 180^\circ$ we can show that

- ☐ $\overline{RD} \parallel \overline{PQ}$
- ☐ $\overline{RP} \parallel \overline{DQ}$

and since $m\angle D + m\angle Q = 180^\circ$ we can show that

- ☐ $\overline{RD} \parallel \overline{PQ}$
- ☐ $\overline{RP} \parallel \overline{DQ}$

by the converse of the

- ☐ alternate interior angles
- ☐ consecutive interior angles

theorem. Therefore, $RPQD$ is a

- ☐ parallelogram
- ☐ kite

by

definition.



After students answer question 2 prior to moving to the next component, clarify any questions or misunderstandings students have about rectangles, before moving to the next component.

8.3.4 Guided Instruction: More Special Parallelograms

Component Overview: Students will measure various segments and angles to decide what properties apply to squares and to rhombi.



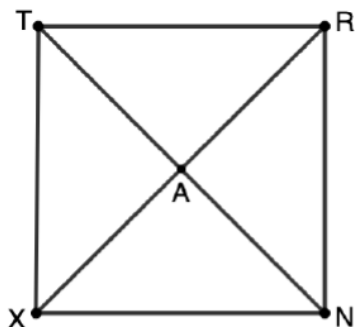
8.3.4 Guided Instruction: More Special Parallelograms



A **square** is a quadrilateral with four right angles and four equal-length sides. It's a member of the following shape classes: polygons, quadrilaterals, trapezoids, parallelograms, rectangles, rhombi.

1. $TRNX$ is a square with diagonals $\overline{TN}$ and $\overline{XR}$ intersecting at point A .

- a. Use a ruler and a protractor to fill in the chart with the appropriate measurements. Include the units.
Lengths depend on student lesson format.



Segment	Measure	Segment/Angle	Measure
$\overline{TN}$	6.6 cm	$\overline{XA}$	3.3 cm
$\overline{XR}$	6.6 cm	$\overline{RA}$	3.3 cm
$\overline{TA}$	3.3 cm	$\angle XAT$	90°
$\overline{NA}$	3.3 cm	$\angle NAR$	90°

- b. Which of the following describe the diagonals of a square? Circle all that apply.

congruent

bisect

perpendicular



If you have limited rulers, allow students to work in pairs to measure the segments, but ask that students take turns measuring.

8.3.4 Guided Instruction: More Special Parallelograms (continued)

- c. List the features that determine if a parallelogram is a square.

All angles are congruent
Diagonals are congruent
Diagonals are perpendicular
All sides are congruent



Consider having students chart the common features between a square and a parallelogram on chart paper.

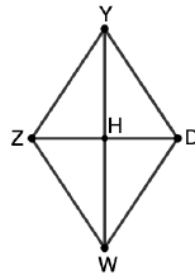


A **rhombus** is a quadrilateral containing four equal-length sides. It's a member of the following shape classes: polygons, quadrilaterals, trapezoids, parallelograms. Squares form a sub-class.

2. $YDWZ$ is a rhombus with diagonals $\overline{YW}$ and $\overline{ZD}$ intersecting at point H .

- a. Use a ruler and a protractor to fill in the chart with the appropriate measurements. Include units.

Lengths depend on student lesson format.



Segment	Measure	Segment/Angle	Measure
$\overline{YW}$	3.6 cm	$\overline{ZH}$	1.2 cm
$\overline{ZD}$	2.4 cm	$\overline{DH}$	1.2 cm
$\overline{YH}$	1.8 cm	$\angle ZHW$	90°
$\overline{WH}$	1.8 cm	$\angle YHD$	90°

- b. Which of the following describe the diagonals of a rhombus? Select all that apply.

congruent

bisect

perpendicular

8.3.4 Guided Instruction: More Special Parallelograms (continued)

- c. List the features that determine if a parallelogram is a rhombus.

Diagonals are perpendicular; All sides are congruent

- d. What are the differences between a square and a rhombus?

In a square, diagonals are congruent. In a rhombus, diagonals may or may not be congruent. A square has all angles equal to 90° . A rhombus has opposite angles equal, though they could be equal to 90° .

3. Use the numbered statements in the box to fill in the Venn diagram comparing rectangles, squares, and rhombi.

1. opposite sides are congruent

2. opposite sides are parallel

3. all angles are congruent

4. diagonals are perpendicular

5. opposite angles are congruent

6. consecutive angles are supplementary

7. all sides are congruent

8. diagonals are congruent

9. diagonals bisect each other

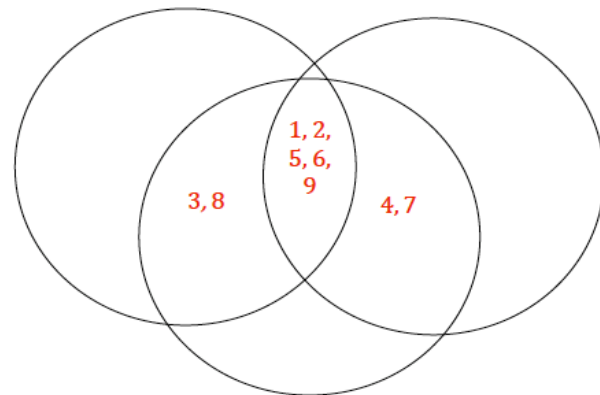


Have students add the common features and differences from question 2 parts c and d to their chart paper. This will assist them later in the lesson when they get to the Venn diagram.

8.3.4 Guided Instruction: More Special Parallelograms (continued)

Rectangle

Rhombus

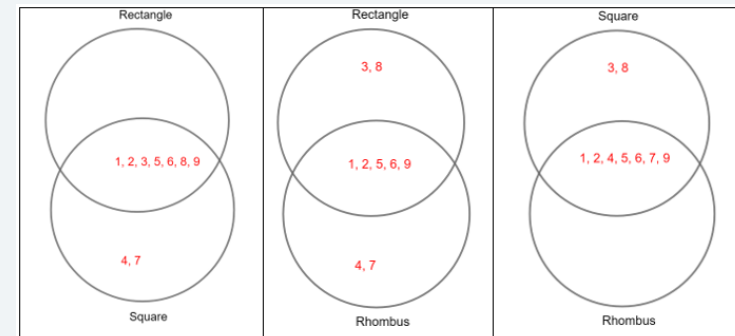


Square



Students may place statements in more than one area of the Venn diagram. For students who struggle with looking at all three figures at one time with the given statements, students can begin by creating a Venn diagram with only two figures:

- Rectangle and Rhombus
- Rectangle and Square
- Rhombus and Square



8.3.5 Bring It Together: Which Special Parallelogram Is It?

Component Overview: Students write statements that classify a figure as either a rectangle, rhombus, or square.



8.3.5 Bring It Together: Which Special Parallelogram Is It?

1. Explain how to show a parallelogram is a rectangle but not a rhombus or a square.

Answers will vary. A parallelogram is a rectangle but not a square or rhombus when it does not have four congruent sides.

2. Explain how to show a parallelogram is a rhombus but not a square or a rectangle.

Answers will vary. A parallelogram is a rhombus but not a square or rectangle when the diagonals are not congruent.

3. Explain how to show a parallelogram is a square.

Answers will vary. A parallelogram is square when diagonals are congruent and all sides are congruent.



Stronger and Clearer Each Time

Consider using this routine as students write explanations in questions 1-3. Have students think and write individually first. Then, through a structured pairing strategy, present the students multiple opportunities to refine and clarify their response through conversation. Finally, students revise their original written response and prepare to share the response with the whole group.

8.3.6 Wrap-Up: Reflection

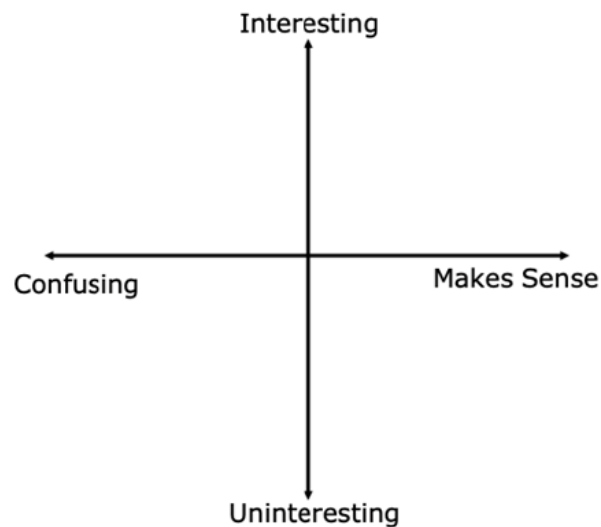
Component Overview: Students will place a mark on the grid that summarizes their comfort level and interest in the learning target from the lesson, and explain why they placed a point at that location.



8.3.6 Wrap-Up: Reflection

1. Plot a point on the grid to indicate your level of interest and how much sense you made of today's learning targets.

Answers will vary.



2. Explain why you placed the point where you did.

Answers will vary.



Monitor for students who mark that they were confused. Make note of those students' explanations to create a plan for addressing their confusions during the next lesson.

8.4 Kites and Parallelograms

Lesson Introduction

 Benchmark	MA.912.GR.1.4 Prove relationships and theorems about parallelograms. Solve mathematical and real-world problems involving postulates, relationships, and theorems of parallelograms.		 Learning Target	I can use properties of parallelograms, kites, and special parallelograms to solve mathematical and real-world problems.
 Benchmark Coherence	Building On		Working Toward	
	N/A		N/A	
 Essential Questions	- What is a kite? - Is a kite a special type of parallelogram?			
 Lesson Narrative	Building on the properties of a parallelogram, in this lesson, students will explore the properties of kites. Students will discover that kites are not a type of parallelogram after classifying descriptions of sides, angles, and diagonals for different quadrilaterals. Using the properties of parallelograms and kites, students will solve mathematical and real-world problems.			
 Component Summary	8.4.1 Warm-Up (~5 min.)	Estimate the number of cans in an artistic structure. (SE p. 26)	8.4.4 Guided Instruction (10-15 min.)	Solve problems using the properties of kites and parallelograms. (SE p. 29)
	8.4.2 Exploration (5-10 min.)	Determine the properties of a kite. (SE p. 26)	8.4.5 Your Turn! (5-10 min.)	Solve problems using the properties of kites and parallelograms. (SE p. 31)
	8.4.3 Collaboration (5-10 min.)	Collaborate to categorize features of different quadrilaterals. (SE p. 28)	8.4.6 Wrap-Up (~5 min.)	Solve a mathematical problem using the properties of a kite. (SE p. 32)
 Resources	Glossary		Materials	
	<u>Key Terms:</u> - Kite <u>Additional Terms:</u> - Quadrilateral - Rectangle - Square - Rhombus		- Ruler - Protractor	
			Additional Preparation	
			Consider making copies of the statements in 8.4.3 Collaboration for students to sort into categories.	

 Teacher Tips	Common Misconceptions • Students may set the wrong segments equal when creating equations. • Students who struggle with fractions may have trouble with 8.4.6 Wrap-Up. Have students show all their steps, so that it can easily be identified if there are issues with the content in the lesson (so you can create a plan for remediation), or if there are calculation errors.	Things to Consider • If time is an issue, then consider assigning 8.4.5 Your Turn! for homework.
	Literacy and Language Routines	Mathematical Thinking and Reasoning (MTR) Standard
	Notice and Wonder 8.4.1 Warm-Up is an opportunity to inspect a unique figure known as a “canstruction.” Students are encouraged to reflect on their wonderings of the image. Consider extending this to asking what they notice, prior to answering question 2.	MA.K12.MTR.7.1: Real-World Contexts Students are provided a variety of real-world contexts to apply the properties of quadrilaterals they have learning thus far in the unit. In both 8.4.4 Guided Instruction and 8.4.5 Your Turn!, students use the properties of parallelograms, rectangles, and kites to solve mathematical and real-world problems.
 Differentiate Instruction	Consider allowing students the option to complete 8.4.2 Exploration in pairs or individually.	
	Consider modeling for students how to mark the text and diagram while reading the situation in 8.4.4 Guided Instruction, question 3, including thinking aloud about why you’re making those markings. Release to students by reading aloud question 2 in 8.4.5 Your Turn!, while students mark the text and diagram.	
Lesson Notes:		

8.4.1 Warm-Up: Canstruction

Component Overview: Students will read through a context about “Canstructions,” and respond to the prompts of what they wonder and how many cans are used to build the structure.



8.4.1 Warm-Up: Canstruction

Each year just before Thanksgiving there are competitions held all over the world to build unique Canstructions with canned food that is collected from the community. The food is then given to local food banks. The picture shows the Canstruction called “Downside Up,” which was the winning exhibit from Canstruction 2010 in New York City.



1. What do you wonder about “Downside Up”?
Answers will vary. I wonder if there is any reinforcing material, such as glue, holding it together.
2. Make an educated guess at the number of cans used to build “Downside Up”. Show your work.
Answers will vary. I guess that there will be 2,508 cans. There appears to be 4 congruent pyramid structures. The square of the level number is the number of cans on that level in one pyramid. The 1st level has 1 can, the 2nd level has 4 cans, the 3rd has 9, and so on. Add the cans on all 12 levels then multiply that by 4 pyramids to get 2,600 cans. Then, it appears on the top level that each pyramid shares a row with the neighboring pyramid which accounts for sharing 92 cans. Subtracting those gives 2,508 cans.
3. Compare your educated guess with your partner’s. Describe how close your guess was to your partner’s guess.
Answers will vary.



Consider utilizing the Notice and Wonder Routine and have students make note of something they notice in the picture, before moving to question 2.



Students may want to explore if more of these exist. If time permits, consider showing students more “canstruction” creations.

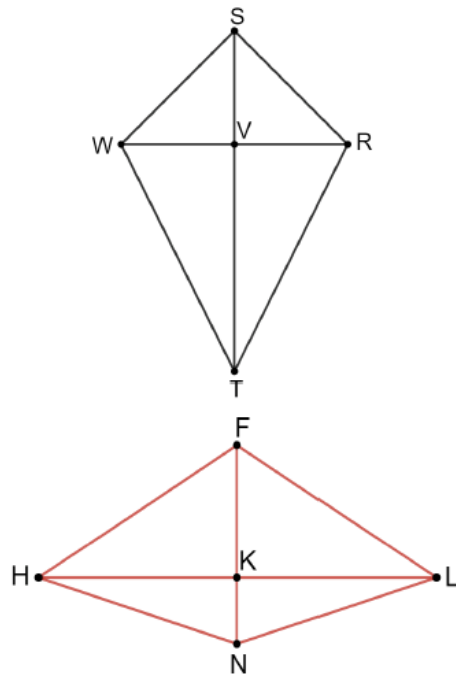
8.4.2 Exploration: What Is a Kite?

Component Overview: Students work collaboratively to measure sides, diagonal lengths, and angles of two kites to determine the properties of this type of quadrilateral.



8.4.2 Exploration: What is a Kite?

Kite $SRTW$ with diagonals $\overline{ST}$ and $\overline{WR}$ intersecting at point V and $FLNH$ with diagonals $\overline{FN}$ and $\overline{HL}$ intersecting at point K are shown.



- Using a ruler, work with a partner to fill in the chart below with the appropriate measurements. Each partner should find the measurements for one kite and share their results with the other.



This Exploration is scaffolded to allow students to engage with it with their partner, rather than being guided through by the teacher. Students may revise their thinking throughout the Exploration as they engage with each new reflection.

Observe and listen as students work with their partners. Make note of students' ideas so that you can ask them to share with the whole group before moving to the next component.

Additional scaffolding questions are provided, if needed, but this should be a primarily student-directed Exploration.



*In looking at the two kites, are they alike in any way(s)? Explain your thinking.
Do you think either of these shapes fit into the definition of quadrilateral, parallelogram, rectangle, square, or rhombus?*

8.4.2 Exploration: What Is a Kite? (continued)

Measurements depend on student lesson format.

<i>SRTW</i>		<i>FLNH</i>	
Segment	Measurement	Segment	Measurement
$\overline{SR}$	4 cm	$\overline{FL}$	5.1 cm
$\overline{SW}$	4 cm	$\overline{FH}$	5.1 cm
$\overline{RT}$	6.3 cm	$\overline{HN}$	4.45 cm
$\overline{WT}$	6.3 cm	$\overline{LN}$	4.45 cm
$\overline{ST}$	8.4 cm	$\overline{FN}$	4.2 cm
$\overline{SV}$	2.8 cm	$\overline{FK}$	2.8 cm
$\overline{TV}$	5.6 cm	$\overline{NK}$	1.4 cm
$\overline{WR}$	5.6 cm	$\overline{HL}$	8.4 cm
$\overline{WV}$	2.8 cm	$\overline{HK}$	4.2 cm
$\overline{RV}$	2.8 cm	$\overline{LK}$	4.2 cm

- a. What do you notice about $\overline{SR}$ and $\overline{SW}$ in *SRTW* and $\overline{FL}$ and $\overline{FH}$ in *FLNH*?

They are congruent.

- b. What do you notice about $\overline{RT}$ and $\overline{WT}$ in *SRTW* and $\overline{HN}$ and $\overline{LN}$ in *FLNH*?

They are congruent.

2. These segment pairs are **adjacent** sides of a kite.

- a. Make a conjecture about the measures of adjacent sides of a kite.

Two sets of adjacent sides of a kite are congruent.

- b. Make a conjecture about the lengths of diagonals of a kite.

The lengths of the diagonals are not congruent. One diagonal bisects the other (but not vice versa).



While the answer key is in inches, answers in other units, such as centimeters, should be accepted as well.



While monitoring students writing their conjectures, listen for accuracy in their reasoning and wording.

Consider having students mark on the kites from question 1, what they note in their conjectures in question 2.

8.4.2 Exploration: What Is a Kite? (continued)

3. Using a protractor, work with your partner to fill in the chart below with the appropriate measurements. Each partner should find the measurements for one kite and share their results with the other.

<i>SRXW</i>		<i>FLNH</i>	
Angle	Measurement	Angle	Measurement
$\angle SVR$	90°	$\angle FKL$	90°
$\angle WSR$	90°	$\angle HFL$	110°
$\angle RTW$	50°	$\angle LNH$	140°
$\angle TWS$	110°	$\angle FHN$	55°
$\angle TRS$	110°	$\angle FLN$	55°

- a. Make a conjecture about what happens when the diagonals of a kite intersect.

When diagonals intersect, they form a 90° and one diagonal is bisected but the other is not

- b. Make a conjecture about the measures of opposite angles of a kite.

One pair of opposite angles of a kite are congruent.

A **kite** is a quadrilateral with exactly two pairs of adjacent congruent sides.



- A **kite** has the following properties:
- One pair of opposite congruent angles
 - Perpendicular diagonals
 - One bisected diagonal



If making a chart, or if students have a foldable about quadrilaterals, have students add this information about kites.

8.4.3 Collaboration: Quadrilaterals

Component Overview: Students work collaboratively to match statements about quadrilaterals to the type(s) of quadrilateral they represent. Once finished, students will compare their matches with other groups, and decide if a kite can be classified as a parallelogram.



8.4.3 Collaboration: Quadrilaterals

1. With your partner, use the bank of phrases to fill in the chart comparing quadrilaterals. Statements can be used more than once.

A. Opposite sides are congruent.	B. Opposite sides are parallel.
C. All sides are congruent.	D. Diagonals bisect each other.
E. Opposite angles are congruent.	F. Diagonals are congruent.
G. All angles are congruent.	H. Diagonals are perpendicular.
I. Two pairs of adjacent sides are congruent.	J. One diagonal is bisected.
K. One pair of opposite angles are congruent.	L. Opposite angles are supplementary.
M. One pair of parallel sides	N. Non-parallel sides are congruent.
O. Two pairs of congruent angles	

a. Write the letter(s) from the bank.

Quadrilateral	Sides	Angles	Diagonals
Parallelogram	A, B	E, O	D
Rectangle	A, B	E, G, L, O	D, F
Square	A, B, C, N, I	E, G, L, O	D, F, H
Rhombus	A, B, C, N, I	E, O	D, H
Kite	I	K	H, J

b. Compare your chart with that of another pair. Add in anything you may have missed.

Answers will vary.

c. Discuss with your partner whether or not kites can be classified as parallelograms.

Answers will vary.



For students who are struggling with identifying all quadrilaterals for a given statement, consider copying the statements to sorting cards, making duplicates for the statements that repeat. For instance, make four copies of statement A, four copies of statement B, two copies of statement C, etc.



Inform students that all phrases may not be used.



Before moving to the next component, address any questions that arose during the Collaboration.

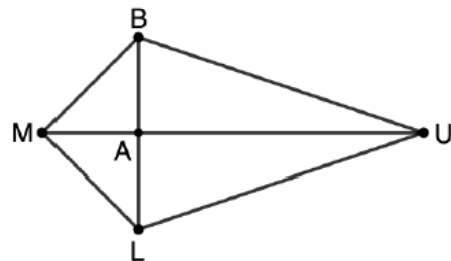
8.4.4 Guided Instruction: Quadrilaterals and Algebra

Component Overview: Students will be guided through using the properties of parallelograms and kites to solve mathematical and real-world problems.



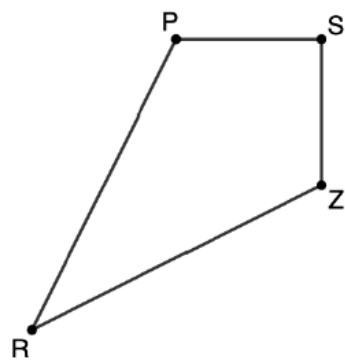
8.4.4 Guided Instruction: Quadrilaterals and Algebra

1. $MBUL$ with diagonals $\overline{BL}$ and $\overline{MU}$ intersecting at point A is shown. Find the value of x that makes $MBUL$ a kite if $BA = 3x - 1$, $LA = 17$, $MA = 2x$, and $UA = 23$.



$$\begin{aligned} BA &= LA \\ 3x - 1 &= 17 \\ x &= 6 \end{aligned}$$

2. Find the values of x and y that make $PSZR$ a kite if $PS = 5x - 4$, $PR = 8y - 4$, $SZ = 3y + 1$, and $RZ = 36$.



$$\begin{aligned} PR &= RZ \\ 8y - 4 &= 36 \\ y &= 5 \end{aligned}$$

$$\begin{aligned} PS &= SZ \\ 5x - 4 &= 3y + 1 \\ 5x - 4 &= 3(5) + 1 \\ x &= 4 \end{aligned}$$

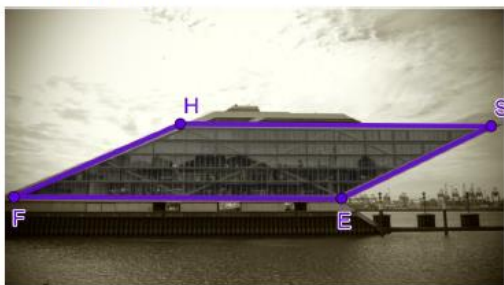


Consider asking the following for question 1, prior to solving for the answer:

- What segments are we given information about?
- What property about kites relate these segments?
- Are both diagonals being bisected?
- Which diagonal is being bisected?
- What does it mean to be bisected?

8.4.4 Guided Instruction: Quadrilaterals and Algebra (continued)

3. The Dockland office building is one of the most famous buildings in Hamburg, Germany. Built in 2005, this building houses roughly 9,000 square meters of office space and breathtaking views of the Elbe River, but the size and the views are not what makes this building so famous. The building itself is in the shape of a parallelogram, $HSEF$.



The perimeter of the Dockland building measures approximately 238 meters. If the length of $\overline{FE}$ is 2.8 times the length of $\overline{FH}$, what are the lengths of $\overline{HS}$, $\overline{SE}$, $\overline{FE}$, and $\overline{FH}$? Round your answers to the nearest thousandth of a meter.

$$FH + SE + HS + FE = 238 \text{ m}$$

$$x + x + 2.8x + 2.8x = 238$$

$$x = 31\frac{6}{19}$$

$$FH \text{ and } SE = 31.316 \text{ m}$$

$$HS \text{ and } FE = 87.684 \text{ m}$$



For question 3, model for students how to mark the text and diagram while reading the situation, including thinking aloud about why you're making those markings. It is important for students to hear the thinking process for this strategy, while also seeing it in action. Marking the text provides students the opportunity to make better sense of the context, while also providing insight for the teacher about what information students are pulling from the text, and what they may be missing.

8.4.5 Your Turn!

Component Overview: Students use the properties of parallelograms to determine if a fence for a given context is a rectangle. Students will also use the properties of kites to solve a real-world problem.



8.4.5 Your Turn!

1. Oliver's carpentry company, Wood 4 You, has been hired to build a fence around a vegetable garden. The gate must be a perfect rectangle. The gate Oliver built is shown, where quadrilateral $BYRX$ outlines the gate.



Oliver wants to be absolutely sure that the gate is a rectangle. Describe two different ways Oliver can use his tape measure to ensure the gate is a true rectangle.

Answers will vary.

He could measure to make sure the diagonals were congruent and are bisecting each other.

From one angle, he could make a mark 3 inches (in.) from the vertex on one leg and 4 in. from the same vertex on the other leg of the angle. If he measures the length between these two marks and it is 5 in., then he knows that angle is a right angle. He can then check to make sure the other 3 angles are right angles using the same method.



If time is an issue, then consider assigning Your Turn! for homework.

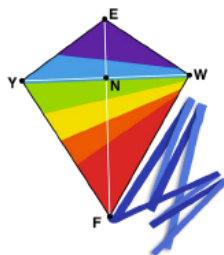


Consider allowing students an alternative to writing a description. Some students may wish to verbally explain to a partner. Have one partner share one of the ways they listed. Their partner can then summarize what they heard, and share a different way to use the tape measure to ensure the gate is a true rectangle.

8.4.5 Your Turn! (continued)

2. Piper is entering a kite-making contest. The kite she designed is shown, where the kite has been labeled as $YEW\!F$ with measurements $EW = 7.5$, $YN = 6$, and $FN = 10.2$, given in inches.

Piper wishes to add a strip of glitter fabric along diagonals $\overline{EF}$ and $\overline{YW}$ that will overlap at their intersection point N , as well as along the perimeter of the kite. How many inches of glitter fabric are needed to cover the diagonals and the perimeter of the kite?



$$\begin{aligned}
 EN &= \sqrt{7.5^2 - 6^2} = 4.5 & EF &= EN + NF = 14.7 \\
 WF &= YF = \sqrt{6^2 + 10.2^2} \approx 11.834 \\
 \text{Amount of fabric} &= YE + EW + WF + YF + YW + EF \\
 \text{Amount of fabric} &\approx 7.5 + 7.5 + 11.834 + 11.834 + 12 + 14.7 \\
 \text{Amount of fabric} &\approx 65.368 \text{ inches}
 \end{aligned}$$



Determine from observational data during the lesson, if a small, teacher-facilitated group is needed to work through question 2.

Another option, consider reading the situation in question 2 aloud while students mark the text and diagram on their own. Then, students could work in pairs to discuss and compare what information they marked, and what they understand about the context, before considering the solution.

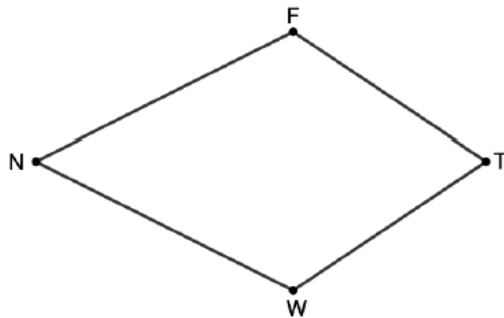
8.4.6 Wrap-Up: Cool Down

Component Overview: Students use the properties of kites to solve for an unknown variable.



8.4.6 Wrap-Up: Cool Down

- Kite $FTWN$ is shown. Find the value of x if $m\angle N = 53^\circ$, $m\angle T = 67^\circ$, and $m\angle F = \left(\frac{1}{2}x - 8\right)^\circ$.



$$m\angle N + m\angle T + m\angle F + m\angle W = 360^\circ$$

$$53^\circ + 67^\circ + \left(\frac{1}{2}x - 8\right)^\circ + \left(\frac{1}{2}x - 8\right)^\circ = 360^\circ$$

$$x + 104 = 360$$

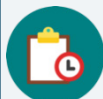
$$x = 256$$



The information provided from the student here, coupled with what was observed during instruction, can provide information for differentiating instruction in later lessons.

8.5 Trapezoids

Lesson Introduction

 Benchmarks	MA.912.GR.1.5 Prove relationships and theorems about trapezoids. Solve mathematical and real-world problems involving postulates, relationships, and theorems of trapezoids. MA.912.LT.4.10 Judge the validity of arguments and give counterexamples to disprove statements.		 Learning Targets	- I can prove relationships and theorems about trapezoids. - I can use relationships and theorems about trapezoids.
 Benchmark Coherence	Building On N/A		Working Toward N/A	
 Essential Questions	- What are the properties of a trapezoid? - What are the properties of an isosceles trapezoid? - How can I prove relationships and theorems about trapezoids, and use them to solve problems?			
 Lesson Narrative	This lesson introduces students to another subset of quadrilaterals called trapezoids. Students will learn the parts of a trapezoid (bases, base angles, and legs) and explore features of trapezoids to write conjectures. Students will apply their knowledge of trapezoids to prove relationships and theorems about trapezoids and use relationships to solve problems.			
 Component Summary	8.5.1 Warm-Up (~5 min.)	Identify and define trapezoids and trapeziums. (SE p. 33)	8.5.4 Guided Instruction (10-15 min.)	Prove theorems about trapezoids. (SE p. 36)
	8.5.2 Collaboration (10 min.)	Determine the properties of isosceles trapezoids. (SE p. 33)	8.5.5 Guided Practice (10 min.)	Solve problems using the properties of trapezoids. (SE p. 38)
	8.5.3 Exploration (5-10 min.)	Name the parts of trapezoids. (SE p. 35)	8.5.6 Wrap-Up (~5 min.)	Solve problems using the properties of trapezoids. (SE p. 38)
 Resources	Glossary		Materials	
	<u>Key Terms:</u> - Trapezoid - Isosceles Trapezoid - Median - Midsegment - Altitude <u>Additional Terms:</u> - Base Angles - Bases - Legs		- Patty Paper - Rulers - Protractors	
			Additional Preparation	
			N/A	

 Teacher Tips	Common Misconceptions	Things to Consider
	Students may use the wrong base angles for 8.5.4 Guided Instruction, question 3.	Consider creating anchor charts of the features of trapezoids and isosceles trapezoids.
	Literacy and Language Routines	Mathematical Thinking and Reasoning (MTR) Standard
	Three Reads Consider the Three Reads routine for the 8.5.1 Warm-Up. This routine supports reading comprehension of problems and meta-awareness of mathematical language. It also supports negotiating information in a text with a partner in mathematical conversation.	MA.K12.MTR.4.1: Discussion and Reflection As students progress through 8.5.2 Collaboration, they are challenged to evaluate a variety of statements. Once the statements are determined to be incorrect, they will discuss writing a new conjecture with their partner. Students will also have the opportunity to compare and discuss their conjectures with another pair of partners.
 Differentiate Instruction	For 8.5.4 Guided Instruction, consider implementing a silent board game. Display the proof, and have students volunteer to go to the board and fill in one of the missing statements or reasons. There is no student talk during this activity. This routine offers students entry into the problem where they feel confident.	
Lesson Notes:		

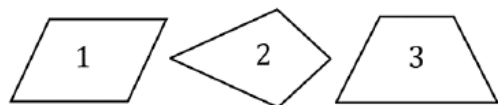
8.5.1 Warm-Up: Trapezoid or Trapezium?

Component Overview: Students review the American and British definitions of Trapezoid and Trapezium, and identify figures that fit each definition.



8.5.1 Warm-Up: Trapezoid or Trapezium?

1. Which quadrilateral is classified as a trapezoid?



There is more than one definition of a trapezoid.

Inclusive Definition	Exclusive Definition
A quadrilateral with at least one pair of parallel opposite sides.	A quadrilateral with exactly one pair of parallel opposite sides.

There is more than one definition of trapezium.

American Definition	British Definition
A quadrilateral with no parallel sides.	A quadrilateral with two sides parallel.

2. Use the definitions to classify quadrilaterals 1, 2, and 3. Write the numbers of the quadrilateral in the table.

Inclusive Definition of Trapezoid	Exclusive Definition of Trapezoid
1, 3	3
American Definition of Trapezium	British Definition of Trapezium
2	1, 3



Three Reads

These definitions have minute differences that can be easily missed. Consider implementing this routine to ensure students notice these differences:

- Students read through the definitions silently.
- Students read the definitions again silently, and make notes and underline important words.
- Students read one definition to their partner, and summarize in their own words what it is saying. Then, alternate who reads on each definition.

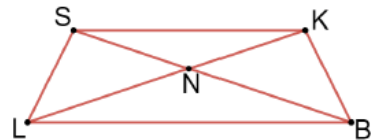
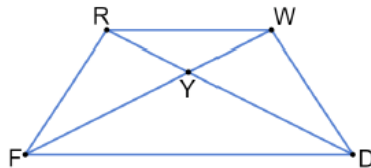
8.5.2 Collaboration: Properties of Isosceles Trapezoids

Component Overview: In this collaborative activity students will measure segment lengths and angles of two isosceles trapezoids to determine if a statement is true or false. If not true, students will provide a conjecture that would be true.



8.5.2 Collaboration: Properties of Isosceles Trapezoids

1. $RWDF$ and $SKBL$ are both **isosceles trapezoids**. $\overline{RD}$ and $\overline{WF}$ are the diagonals of $RWDF$ and intersect at point Y . $\overline{SB}$ and $\overline{KL}$ are the diagonals of $SKBL$ and intersect at point N .



- a. Using a ruler and a protractor to measure $RWDF$ and $SKBL$, work with your partner to determine if the following conjectures about isosceles trapezoids are true or false. If false, write a new conjecture based on your measurements.



Encourage students to measure and organize their measurements prior to moving on to the table. Encourage students to choose whether they will split the work of measuring, or both measure everything and then compare. Allowing students to develop their own method of splitting the work teaches valuable cooperative and collaborative skills that can be used in the real world.

8.5.2 Collaboration: Properties of Isosceles Trapezoids (continued)

Conjecture	True or False?	If false, write a new conjecture.
All angles of an isosceles trapezoid are congruent	<i>False</i>	<i>Isosceles trapezoids have 2 pairs of congruent angles.</i>
Opposite sides are parallel	<i>False</i>	<i>One set of opposite sides are parallel.</i>
Opposite angles are congruent	<i>False</i>	<i>Opposite angles of an isosceles trapezoid are supplementary.</i>
Opposite sides are congruent	<i>False</i>	<i>One pair of opposite sides of an isosceles trapezoid are congruent.</i>
Diagonals bisect each other	<i>False</i>	<i>Diagonals of an isosceles trapezoid are congruent but do not bisect each other.</i>



Students sometimes choose to pick the answer that requires the less work, not because it is the correct answer. To avoid this, for the conjectures that students mark as true, set the expectation that they must present a drawing that supports the truth of the conjecture.

- b. Compare your chart to that of another pair. Discuss any differences. Make corrections as necessary.

Answers will vary.

2. $\angle RFD$ and $\angle WDF$ in $RWDF$ are called **base angles**. $\angle SLB$ and $\angle KBL$ are base angles in $SKBL$.

Based on your previous measurements, make a conjecture about the base angles of an isosceles trapezoid.

Base angles are congruent.

3. Each isosceles trapezoid has two pairs of base angles.
- What is the other pair of base angles in trapezoid $RWDF$?
 - What is the other pair of base angles in trapezoid $SKBL$?

$\angle FRW$ and $\angle DWR$ are the other base angles.

$\angle LSK$ and $\angle BKS$ are the other base angles.



MA.K12.MTR.4.1: Discussion and Reflection
Students are encouraged to discuss and reflect with another pair of partners. Students should be encouraged to take risks and offer statements that others may disagree with. It is important as a class to meaningfully consider all opinions.

8.5.2 Collaboration: Properties of Isosceles Trapezoids (continued)



A **trapezoid** is a quadrilateral with *at least* one pair of parallel sides. It's a member of the following shape classes: polygons, quadrilaterals. Its sub-classes include parallelograms, rectangles, rhombuses, and squares.

An **isosceles trapezoid** is a trapezoid with two pairs of congruent base angles.

4. Make a conjecture about the legs of an isosceles trapezoid.

Answers will vary. The legs of an isosceles trapezoid will be congruent.



If an isosceles trapezoid has at least one pair of congruent base angles, what is true of the side lengths of an isosceles trapezoid?

Answer: You can use AAS to prove that opposite sides are congruent.

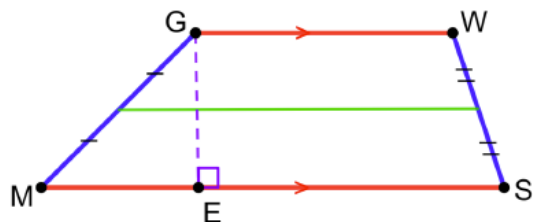
8.5.3 Exploration: Parts of a Trapezoid

Component Overview: Students mark and explore the parts of a trapezoid to determine information that is true for a trapezoid.



8.5.3 Exploration: Parts of a Trapezoid

1. Trapezoid $GWSM$ is shown.



- a. The parallel sides of a trapezoid are called the **bases**. The trapezoid shown has one pair of bases.

Label each base red and mark them to show they are parallel on $GWSM$.

The bases have been marked on the trapezoid in red.

- b. The non-parallel sides of a trapezoid are called the **legs**. The trapezoid shown has two legs.

Label each leg of $GWSM$.

The legs have been marked on the trapezoid in blue.



The **median** of a trapezoid connects the non-parallel sides at their midpoints. The median is sometimes referred to as the **midsegment**.

- c. Using your ruler, plot the midpoint of $\overline{GM}$ and the midpoint of $\overline{WS}$. Connect these points to create the median of trapezoid $GWSM$ in green. Be sure to mark the legs appropriately showing that the median is connected at the midpoints.

The midpoints have been connected with a green line.



Consider creating an anchor chart with the parts of a trapezoid, and at the conclusion of 8.5.3 Exploration, add any pertinent information learned from question 2.

8.5.3 Exploration: Parts of a Trapezoid (continued)



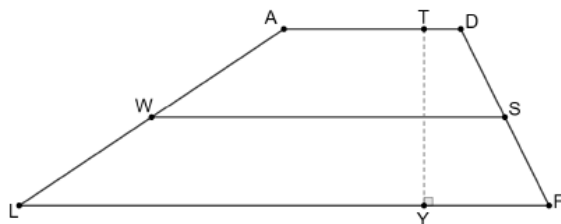
An **altitude** of a trapezoid is any segment from a point on one of the bases perpendicular to the line containing the other base.

- d. The length of the altitude is the distance between the bases.

Draw the altitude on $GWSM$. Name the altitude by naming the endpoints and mark the trapezoid to show that the altitude is perpendicular to the base.

The altitude has been indicated on the trapezoid with a dashed purple line and is labeled $\overline{GE}$.

2. Trapezoid $ADFL$ is shown with midsegment $\overline{WS}$ and altitude $\overline{TY}$.



Select all of the information that is true about $ADFL$.

- | | | |
|--|--|--|
| ☐ $\overline{DF} \cong \overline{AL}$ | ☐ $m\angle LAD = m\angle FDA$ | ☐ $\overline{TY} \perp \overline{WS}$ |
| ☐ $\overline{AD} \parallel \overline{FL}$ | ☐ $\overline{WS} \parallel \overline{FL}$ | ☐ $m\angle AWS = m\angle DSW$ |
| ☐ $\overline{TY} \perp \overline{AD}$ | ☐ $\overline{DS} \cong \overline{SF}$ | ☐ $m\angle SFL = m\angle DSW$ |



This would be a great opportunity to do a “Live poll” using any “Voting” tool you like. Offer each option as a separate question for students to respond Yes, No, or I do not know. This data can be used as a formative assessment, if you count the number of each response for each question.

- Count the number of incorrect or “I do not know” answers, and decide if this is too many to move on.
- Consider adding other possible statements to the poll to get a feel for the class.

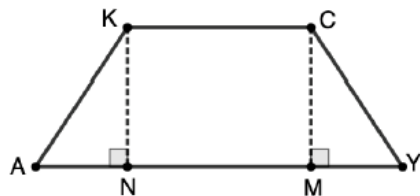
8.5.4 Guided Instruction: Proving Isosceles Trapezoid Properties

Component Overview: Students will use patty paper to determine properties and develop theorems of isosceles trapezoids. Then, using the properties, guide the students through a two-column and a paragraph proof that will prove the theorems.



8.5.4 Guided Instruction: Proving Isosceles Trapezoid Properties

1. Trapezoid $KCYA$ is shown with altitudes $\overline{KN}$ and $\overline{CM}$.

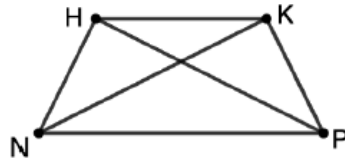


- Trace the trapezoid onto a piece of patty paper.
- Fold the patty paper so that vertices K and C coincide.
- Describe what else you notice when you fold the patty paper.
Folding the patty paper created two congruent figures.
- Work with a partner to list all of the congruent parts of trapezoid $KCYA$.
 $\overline{KA} \cong \overline{CY}$, $\angle KAN \cong \angle CYM$, $\angle AKC \cong \angle YCK$
- Compare your list with that of another pair. Add any congruent parts you may have missed.
Answers will vary.
- Describe how the reflecting across the midpoint of $\overline{KC}$ shows that $KCYA$ can be classified as an isosceles trapezoid.
An isosceles trapezoid has congruent base angles. The reflection shows this trapezoid has two pairs of base angles congruent.

8.5.4 Guided Instruction: Proving Isosceles Trapezoid Properties (continued)

2. Given: $HKPN$ is an isosceles trapezoid with $\overline{HK} \parallel \overline{PN}$.

Prove: $\overline{HP} \cong \overline{KN}$



Complete the following two-column proof.

Statement	Reason
1. $HKPN$ is an isosceles trapezoid; $\overline{HK} \parallel \overline{PN}$	1. Given
2. $\overline{HN} \cong \overline{KP}$	2. The legs of an isosceles trapezoid are congruent.
3. $\angle HNP \cong \angle KPN$	3. Base angles of an isosceles trapezoid are congruent
4. $\overline{NP} \cong \overline{NP}$	4. Reflexive Property of Congruence
5. $\triangle HNP \cong \triangle KPN$	5. SAS Congruence Theorem
6. $\overline{HP} \cong \overline{KN}$	6. CPCTC

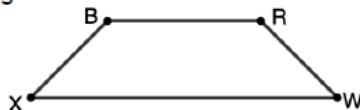


Consider implementing a silent board game for this proof. Display the proof and have students volunteer to go to the board and fill in one of the missing statements or reasons. There is no student talk during this activity. This routine offers students entry into the problem where they feel confident.

8.5.4 Guided Instruction: Proving Isosceles Trapezoid Properties (continued)

3. Given: $BRWX$ is an isosceles trapezoid and $\overline{BR} \parallel \overline{XW}$.

Prove: $\angle RBX$ and $\angle RWX$ are supplementary, $\angle BRW$ and $\angle BXW$ are supplementary.



To add a challenge, provide the question stem, but not the partially completed proof. Have students complete the proof using any form they choose (two-column, paragraph, or flow-proof).

Given that $BRWX$ is an isosceles trapezoid, then

$\angle RBX \cong \angle BRW$ because base angles of an isosceles trapezoid are congruent. Therefore, by the definition of

congruence, $m\angle RBX = m\angle BRW$. Since it is given that

$\overline{BR} \parallel \overline{XW}$, then $\angle RBX$ and $\angle BXW$ are supplementary

and $\angle RWX$ and $\angle BRW$ are supplementary

because when parallel lines are intersected by a

transversal, consecutive interior angles are

supplementary. Therefore, by definition of supplementary

angles, $m\angle BRW + m\angle RWX = 180^\circ$ and

$m\angle RBX + m\angle BXW = 180^\circ$. By substitution,

$m\angle RBX + m\angle RWX = 180^\circ$ and $m\angle BRW + m\angle BXW = 180^\circ$.

Therefore, by definition of supplementary angles,

$\angle RBX$ and $\angle RWX$ are supplementary and $\angle BRW$ and $\angle BXW$

are supplementary.



There are two pairs of congruent base angles. Be aware to state that the only pair that will work with the other statements in this paragraph proof are $\angle RBX$ and $\angle BRW$.

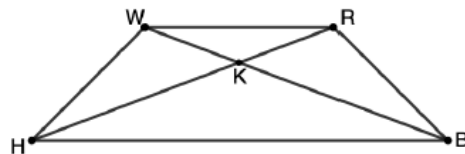
8.5.5 Guided Practice: Isosceles Trapezoids

Component Overview: Students analyze a trapezoid and use properties of parallel lines, triangles, and other shapes to find missing angles and solve for unknown variable values.



8.5.5 Guided Practice: Isosceles Trapezoids

1. $WRBH$ is an isosceles trapezoid.



- a. If diagonal $WB = 18$ and diagonal $RH = 3x - 7$, find the value of x .

$$\begin{aligned} WB &= RH \\ 18 &= 3x - 7 \\ x &= 8\frac{1}{3} \end{aligned}$$

- b. Diagonals $\overline{WB}$ and $\overline{RH}$ intersect at point K . If $m\angle RWB = 20.6^\circ$, $m\angle RHW = 24.4^\circ$, and $m\angle BKH = 138.8^\circ$, what is $m\angle WRB$?

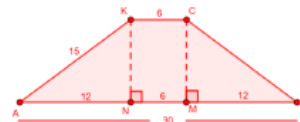
$$\begin{aligned} m\angle WKH &= 180^\circ - m\angle BKH = 180^\circ - 138.8^\circ = 41.2^\circ \\ m\angle HWK &= 180^\circ - m\angle RHW - m\angle WKH \\ &= 180^\circ - 24.4^\circ - 41.2^\circ = 114.4^\circ \\ m\angle HWR &= m\angle HWK + m\angle BWR = 114.4^\circ + 20.6^\circ = 135^\circ \\ m\angle WRB &= m\angle HWR = 135^\circ \end{aligned}$$

2. Isosceles trapezoid $KCYA$ has altitudes $\overline{KN}$ and $\overline{CM}$. If $KC = 6$, $AY = 30$, and $KA = 15$, what is the length of altitude $\overline{CM}$?

The information is sketched and segment addition is used to determine the values of the segments that make up $\overline{AY}$.

Use the Pythagorean theorem to solve for the altitude $\overline{KN}$.

$$\begin{aligned} 15^2 &= 12^2 + (KN)^2 \\ KN &= 9 \\ KN &= CM = 9 \end{aligned}$$



For students who are struggling, consider providing additional scaffolding for question 1 part b:

- If the $m\angle BKH = 38.9^\circ$, then $m\angle WKH = \underline{\hspace{2cm}}^\circ$ because the two angles are .
- $m\angle HWK = \underline{\hspace{2cm}}^\circ$ because all three angles of a triangle equal °.
- $m\angle HWR = m\angle HWK + m\angle BWR = \underline{\hspace{2cm}}^\circ$
- $m\angle HWR = m\angle WRB = \underline{\hspace{2cm}}^\circ$ because the angles of isosceles triangles are congruent and equal.

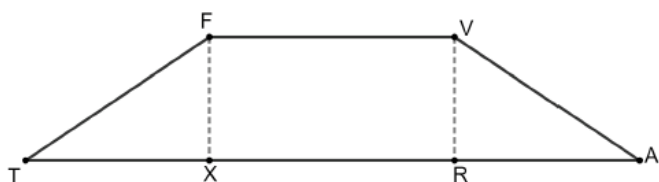
8.5.6 Wrap-Up: Ticket Out the Door

Component Overview: Students use properties of trapezoids to find an unknown length. Students should work alone to demonstrate their knowledge of the properties of trapezoids. Students should not use a calculator to complete the Wrap-Up.



8.5.6 Wrap-Up: Ticket Out the Door

1. Isosceles trapezoid $FVAT$ is shown with altitudes $\overline{FX}$ and $\overline{VR}$.



If $FT = 12$, $TA = 36\sqrt{5}$, and $VR = 8$, what is the length of $\overline{FV}$?

$$FX = VR = 8$$

$$RA = TX$$

Using the Pythagorean Theorem

$$TX = \sqrt{TF^2 - FX^2} = \sqrt{144 - 64} = 4\sqrt{5}$$

$$TA = TX + XR + RA$$

$$36\sqrt{5} = 4\sqrt{5} + XR + 4\sqrt{5}$$

$$XR = 28\sqrt{5}$$

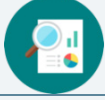
$$FV = XR = 28\sqrt{5}$$

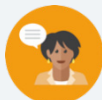


Use the data from 8.5.6 Wrap-Up as a formative assessment. This data can be used to determine students' mastery of the content presented in the lesson, and give information on which, and where, students might need further remediation.

8.6 Solving Real-World Problems With Quadrilaterals

Lesson Introduction

 Benchmarks	MA.912.GR.1.4 Prove relationships and theorems about parallelograms. Solve mathematical and real-world problems involving postulates, relationships, and theorems of parallelograms. MA.912.GR.1.5 Prove relationships and theorems about trapezoids. Solve mathematical and real-world problems involving postulates, relationships, and theorems of trapezoids.		 Learning Targets	- I can use properties of parallelograms to solve real-world problems. - I can use properties of trapezoids to solve real-world problems.
 Benchmark Coherence	Building On N/A		Working Toward N/A	
 Essential Question	- How can the properties of parallelograms be used to solve real-world problems? - How can the properties of trapezoids be used to solve real-world problems?			
 Lesson Narrative	This lesson brings together the properties and relationships learned in Lessons 1-5 about parallelograms and trapezoids to solve real-world problems.			
 Component Summary	8.6.1 Warm-Up (~5 min.)	Explain errors found in student samples. (SE p. 39)	8.6.3 Your Turn! (5-10 min.)	Apply knowledge of quadrilaterals. (SE p. 41)
	8.6.2 Activity (35-40 min.)	Solve real-world problems using parallelograms. (SE p. 40)	8.6.4 Wrap-Up (~5 min.)	Reflect on understanding of the properties covered thus far covered in from the unit. (SE p. 42)
 Resources	Glossary		Materials	Additional Preparation
	<u>Key Terms:</u> - N/A <u>Additional Glossary:</u> - N/A		(Optional) Chart paper (Optional) Markers	- This lesson does include a Gallery Walk that needs to be prepared. Consider making additional sets of the problems for large classes. - If using task cards, consider laminating for use in multiple classes. - Consider strategic grouping for 8.6.2 Activity, prior to the start of the lesson.

 Teacher Tips	Common Misconceptions	Things to Consider
	Students may set the wrong expressions equal when creating equations to solve.	- For 8.6.2 Activity, consider displaying anchor charts that show the classifications of quadrilaterals, or that show the properties covered thus far in the unit. If students have made a foldable, allow them to refer to it during the activity. - If time is an issue, then consider assigning 8.6.3 Your Turn! for homework.
	Literacy and Language Routines	Mathematical Thinking and Reasoning (MTR) Standard
	Take Turns 8.6.2 Activity is a collaborative Gallery Walk. Students should take turns discussing their work to be able to come to a consensus on each answer. Consider having students alternate who is writing the final answer, but each student should take turns explaining their thinking to allow for both students to have a voice, prior to finalizing their collaborative answer.	MA.K12.MTR.7.1: Real-World Contexts 8.6.2 Activity students are expected to apply the properties learned from the unit to real-world contexts. While students may not initially think about quadrilaterals in the real-world when solving problems, this activity will heighten the awareness of how properties and theorems can be used to solve problems for real-world contexts.
 Differentiate Instruction	Use observational data from prior lessons to determine if there needs to be a teacher-facilitated group for the 8.6.2 Activity. If there is a teacher-facilitated group, keep it small and ensure gradual release occurs during the activity to position students to be more confident in the content by the end of the activity.	
Lesson Notes:		

8.6.1 Warm-Up: Error Analysis

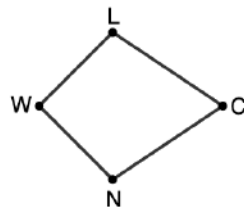
Component Overview: Students review the work of two students that have both made mistakes, and write a letter to each student explaining the mistakes made.



8.6.1 Warm-Up: Error Analysis

Kite $WLCN$ is shown, with $m\angle W = 90^\circ$, $m\angle C = 68^\circ$, and $m\angle N = (6x + 5)^\circ$.

Marcia and Eugene both worked to determine the value of x . Their work is shown.



Marcia	Eugene
$90 + 90 + 68 + 6x + 5 = 360$ $6x + 5 + 248 = 360$ $6x + 253 = 360$ $6x = 107$ $x = \frac{107}{6}$	$360 - (90 + 68) = 202$ $6x + 5 = 202$ $6x = 197$ $x = \frac{197}{6}$



This is an opportunity for students to practice coaching peers, and not just give the answer to students who have made mistakes. Consider allowing students to share their letters and revise based on their partner's feedback.

- Evaluate each student's work, then write a note to each student explaining the error they made.

Marcia	Eugene
<i>Marcia,</i> It looks like you assumed the $m\angle L = 90^\circ$. The measure of $\angle L$ will be equal to $m\angle N$ so you could double that and try to solve with your work = 360°. Try it and let me know what you get!	<i>Eugene,</i> It looks like you forgot the measure of angle $\angle L$ when setting up your equation on the second line. The measure of $\angle L$ will be equal to $m\angle N$ so you could double that and try to solve with your work = 360°. Try it and let me know what you get!

8.6.2 Activity: World Problem Gallery Walk

Component Overview: Students will engage with a series of real-world problems, and decide what properties of parallelograms and trapezoids are needed to solve each. Students will collaborate on their work to come to a consensus on their answers. Circulate among the groups to ensure that one student is not doing all of the work.



8.6.2 Activity: Word Problem Gallery Walk

1. Work through the word problems with your partner. Show your work and write your answers in the appropriate boxes below.

Problem Number	Work	Answer
1	$KR = DH$ $\frac{1}{2}x + 5 = 2x - 4$ $6 = x$ $KR = \frac{1}{2}(6) + 5 = 8 \text{ in.}$	$KR = 8 \text{ in.}$
2	$DW = NF$ $5y - 5 = 2y + 4$ $y = 3$ $DN = WF$ $12x + 4 = 20x - 8$ $x = 1.5$ $\text{Area} = (5(3) - 5)(12(1.5) + 4)$ $\text{Area} = 220 \text{ ft}^2$	$\text{Area} = 220 \text{ ft}^2$
3	$m\angle MGS = m\angle PSG$ $(8x - 10)^\circ = (6x + 12)^\circ$ $x = 11$ $m\angle MGS = (8(11) - 10)^\circ = 78^\circ$ $m\angle PSG = (6(11) + 12)^\circ = 78^\circ$ $m\angle GMP = m\angle MPS$ $(22y - 19)^\circ = (16y + 14)^\circ$ $y = 5.5$ $m\angle GMP = (22(5.5) - 19)^\circ = 102^\circ$ $m\angle MPS = (16(5.5) + 14)^\circ = 102^\circ$	$m\angle MGS = 78^\circ$ $m\angle PSG = 78^\circ$ $m\angle GMP = 102^\circ$ $m\angle MPS = 102^\circ$



Students should take turns explaining how to approach the problem and/or take turns in sharing their work. Then, students should come to a consensus on the correct answer(s).

8.6.2 Activity: World Problem Gallery Walk (continued)

4	$m\angle PCH = m\angle PTH$ $(6x - 23)^\circ = (4x + 19)^\circ$ $x = 21$ $m\angle PCH = (6(21) - 23)^\circ = 103^\circ$ $2(m\angle PCH)^\circ + 2(m\angle CPT)^\circ = 360^\circ$ $2(103)^\circ + 2(8y - 23)^\circ = 360^\circ$ $y = 12.5$	$y = 12.5$
5	$FM = MA$ $3x + 20 = 5x - 3$ $x = 11.5$ $FM = 3(11.5) + 20 = 54.5$ $RM = \frac{1}{2}(FL)$ $RM = \frac{1}{2}(8x - 1)$ $RM = \frac{1}{2}(8(11.5) - 1) = 45.5$ $TM = \frac{1}{2}(FG)$ $TM = \frac{1}{2}(5x + 2.5)$ $TM = \frac{1}{2}(5(11.5) + 2.5) = 30$	$FM = 54.5$ in. $RM = 45.5$ in. $TM = 30$ in.
6	$m\angle DMF + m\angle MDW + m\angle DWF + m\angle WFM = 360^\circ$ $(13x + 4.4)^\circ + (10x + 13)^\circ + (15x + 4)^\circ + (10x - 7)^\circ = 360^\circ$ $x = 7.2$ $m\angle DMF = (13(7.2) + 4.4)^\circ = 98^\circ$ $m\angle DWF = (15(7.2) + 4)^\circ = 112^\circ$ $m\angle MDW = (10(7.2) + 13)^\circ = 85^\circ$ $m\angle WFM = (10(7.2) - 7)^\circ = 65^\circ$ <i>Because opposite angles are not congruent, this is not a parallelogram.</i>	<i>DWFM is not a parallelogram.</i>



Consider one of the following options for this activity.

- **Gallery Walk** – hang problems around the room. Groups are assigned to work one problem on chart paper and post. When all groups have finished, the groups rotate from problem to problem, verifying and editing any mistakes.
- **Task Cards** – print the problems on separate pieces of paper. Give each group one card to complete. When finished, pass the card to the next group. Continue until all cards have been completed.

For large groups, consider making two sets of problems.

8.6.2 Activity: World Problem Gallery Walk (continued)

7	$HX = 5x - 2$ $WR = 2 \cdot HX$ and $WR = 9x + 3$ $9x + 3 = 2(5x - 2)$ $7 = x$ $KR = 11x + 19$ $KR = 11(7) + 19 = 96 \text{ cm}$ $WR = 9(7) + 3 = 66 \text{ cm}$	The dimensions of $KRWF$ are $96 \text{ cm} \times 66 \text{ cm}$
8	$XW = DN$ $12x - 15 = 9x + 27$ $x = 14$ $DW = XN$ $DW = (8x + 40.5) \text{ m}$ $DW = (8(14) + 40.5) \text{ m}$ $DW = 152.5 \text{ m}$	$DW = 152.5 \text{ m}$



When all pairs are finished, the suggested activity is to put pairs in larger groups to compare answers, discuss any differences, and make corrections. Then, larger groups write final answers on chart paper designated for each problem. At that point, discuss any differences as a class, and work together to correct mistakes.

8.6.3 Your Turn!

Component Overview: Students respond to prompts about quadrilaterals that allow them to analyze each type of quadrilateral they have learned about in prior lessons. Students will have to select quadrilaterals that fulfill a condition, then offer a description of a given quadrilateral.



8.6.3 Your Turn!

- I am a quadrilateral with two pairs of opposite parallel sides. At least one of my angles has a measure of 90° and both of my diagonals measure 12 inches. Select all shapes that match this description.
 - ☐ Parallelogram
 - ☐ Rectangle
 - ☐ Square
 - ☐ Rhombus
 - ☐ Kite
- In your own words, write a description of a parallelogram that could **only** match a square.
A square is a quadrilateral with four equal sides, four right angles, and diagonals that are congruent.
- If a quadrilateral is a parallelogram, what additional information would be needed to show that the shape is a rectangle?
To show a parallelogram is a rectangle, it would need to have four right angles and diagonals that are congruent.
- If a quadrilateral is a parallelogram, what additional information would be needed to show that the shape is a rhombus?
To show a parallelogram is a rhombus, it would need to have four congruent sides.



If time is an issue, then consider assigning 8.6.3 Your Turn! for homework.



Consider allowing students options for responding to questions 1-3. For example, instead of students writing their answers, allow them to choose to explain their thinking verbally to a partner who they haven't worked with so far in the lesson.

8.6.4 Wrap-Up: Reflection

Component Overview: Students complete a self-reflection on their level of understanding of properties and completing proofs of parallelograms, rectangles, squares, rhombi, kites, and trapezoids.



8.6.4 Wrap-Up: Reflection

1. Circle the icon that best describes your understanding of each concept below.



I need help.



I got it.



I could help someone.

Answers will vary

Properties of a parallelogram			
Properties of a rectangle			
Properties of a kite			
Completing proofs using properties and theorems about parallelograms			
Using the properties of a parallelogram to solve problems			
Properties of a trapezoid			
Using the properties of a trapezoid to solve problems			

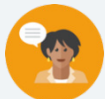
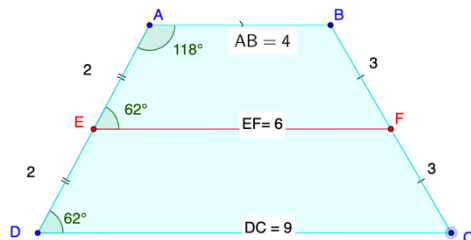


Make note of students who mark any statements as “I need help.” After the lesson, determine a plan for how to provide support to those students (i.e., strategic grouping during the next lesson or during the end of unit review, pull small groups, etc.).

8.7 Trapezoid Midsegment Theorem

Lesson Introduction

 Benchmark	MA.912.GR.1.5 Prove relationships and theorems about trapezoids. Solve mathematical and real-world problems involving postulates, relationships, and theorems of trapezoids.		 Learning Targets	- I can prove the trapezoid midsegment theorem. - I can use properties of trapezoids to solve mathematical and real-world problems.
 Benchmark Coherence	Building On		Working Toward	
	N/A		N/A	
 Essential Questions	- What is the trapezoid midsegment theorem? - How can midsegments be used to find missing lengths of a trapezoid?			
 Lesson Narrative	In this lesson, students will continue their work with trapezoids to discover the Trapezoid Midsegment Theorem. Once students have an opportunity to explore and construct a definition for the Trapezoid Midsegment Theorem, they are given an opportunity to apply the concept theoretically, and in real-world situations.			
 Component Summary	8.7.1 Warm-Up (~5 min.)	Notice and Wonder for trapezoids in architecture. (SE p. 43)	8.7.4 Guided Practice (10 min.)	Find measures using the trapezoid midsegment theorem. (SE p. 46)
	8.7.2 Exploration (5-10 min.)	Explore the relationship between lengths in a trapezoid. (SE p. 43)	8.7.5 Your Turn! (5-10 min.)	Find measures using the trapezoid midsegment theorem. (SE p. 47)
	8.7.3 Guided Instruction (5-10 min.)	Prove the trapezoid midsegment theorem. (SE p. 44)	8.7.6 Wrap-Up (~5 min.)	Summarize the lesson as a note to a friend. (SE p. 48)
 Resources	Glossary		Materials	
	Key Terms: - Midsegment - Trapezoid Midsegment Theorem Additional Glossary: N/A		- Patty paper (Optional) Scissors	
		Additional Preparation N/A		

 Teacher Tips	Common Misconceptions Students may set half the length of the midsegment and one of the bases equal to the other base.	Things to Consider Students will continue the work of the triangle midsegment theorem in a later lesson. For this lesson, students just need a surface understanding of what the midsegment of a triangle is, and how its length is related to the base of the triangle.
	Literacy and Language Routines Notice and Wonder This routine provides a low-floor, high-ceiling approach to create pathways that connect concrete observations to abstract concepts. Each activity in this lesson can use Notice and Wonder, as students can connect what they see about midsegments to their mathematical properties, ultimately formalizing this connection as the Trapezoid Midsegment Theorem.	Mathematical Thinking and Reasoning (MTR) Standard MA.K12.MTR.5.1: Patterns and Structure In the 8.7.2 Exploration activity, students make use of measurements to begin formulating the patterns and connections that ultimately become their working definition of the Trapezoid Midsegment Theorem. Utilize this opportunity to build students' capacity for making use of patterns, rather than memorizing a theorem.
 Differentiate Instruction	Digital applets provide manipulatives for students to explore midsegments without having to constantly erase and draw new shapes. For example, GeoGebra has a trapezoid midsegment applet in which students can adjust the length of the base to predict and verify how the length of the midsegment might change. Consider utilizing digital manipulatives during 8.7.4 Guided Practice or 8.7.5 Your Turn! for additional scaffolding or extending to predict lengths for other shapes.	
Lesson Notes:		

8.7.1 Warm-Up: Trapezoids in Architecture

Component Overview: Students engage in a Notice and Wonder routine to make meaningful connections to trapezoids in the real world.



8.7.1 Warm-Up: Trapezoids in Architecture

Four different buildings are shown.

Machu Picchu, Peru	Greenland Center, Beijing, China
	
Garden Terrace, Nagasaki	International Spy Museum, Washington D.C.
	



Notice and Wonder

Allow students the opportunity to notice and wonder individually and then discuss with a partner the possible reasons the trapezoids were used the way they were.

- What do you notice about the buildings?
Answers will vary. I notice that they all include a trapezoid shape.
- Describe how a trapezoid is used in each building.
Answers will vary. In Machu Picchu, a trapezoid is used as an entryway. The Greenland Center uses inverted trapezoids similar to bricks. Garden Terrace appears to be in the shape of a trapezoid. The International Spy Museum uses trapezoid shapes at one end of the building.

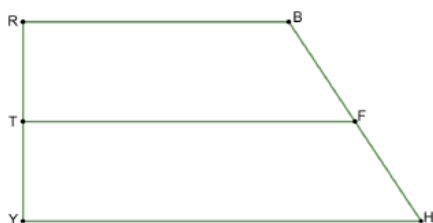
8.7.2 Exploration: Midsegment of a Trapezoid

Component Overview: Students use patty paper to explore visual and mathematical relationships between trapezoids and their midsegments. Students can work in pairs or small groups as the teacher facilitates.



8.7.2 Exploration: Midsegment of a Trapezoid

1. Trapezoid $RBHY$ is shown, where $\overline{RB} \parallel \overline{YH}$.



- a. Use patty paper or a ruler to determine if $\overline{TF}$ is the midsegment of trapezoid $RBHY$. Justify your conclusion.

Point T is the midpoint of $\overline{RY}$ and point F is the midpoint of $\overline{BH}$, therefore $\overline{TF}$ is the midsegment.

- b. Use a ruler to complete the table.

Answers will vary depending on the units used.

Segment(s)	Measure
$\overline{RB}$	$RB = 6.7 \text{ cm}$
$\overline{YH}$	$YH = 10 \text{ cm}$
$\overline{TF}$	$TF = 8.35 \text{ cm}$

2. What do you notice about the relationship between the length of the midsegment and the sum of the lengths of the bases?

The length of the midsegment is equal to half the length of the sum of the bases.

3. Write an equation to show the relationship between the length of the midsegment of a trapezoid and the length of its bases.

$$\text{Midsegment} = \frac{1}{2}(b_1 + b_2)$$



Allow students to choose any unit of measure for this activity. The units themselves do not matter, but the relationship between the lengths does.



Notice and Wonder

Consider implementing this routine to allow students an opportunity to generate ideas about trapezoids. Monitor for student responses that can later be revisited and formalized to make meaningful connections during the proof in 8.7.3 Guided Instruction.

8.7.3 Guided Instruction: Trapezoid Midsegment Theorem

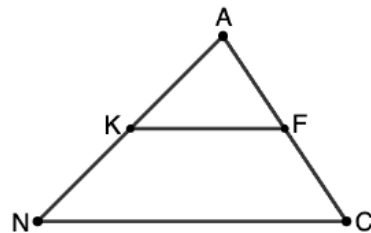
Component Overview: Teacher facilitates a proof that connects different properties of a trapezoid to the Trapezoid Midsegment Theorem.



8.7.3 Guided Instruction: Trapezoid Midsegment Theorem

The **midsegment** of a triangle is a segment joining the midpoints of two sides of a triangle.

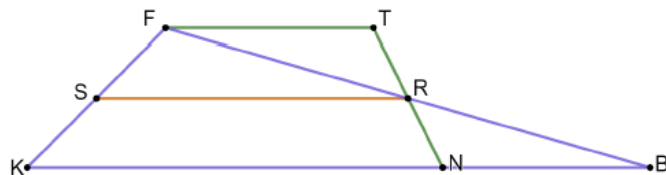
1. $\triangle ACN$ is shown with midsegment $\overline{KF}$.



Fill in the blanks to complete the triangle midsegment theorem.

The triangle midsegment theorem states that the midsegment of a triangle is **parallel** to the third side it does not touch and is **half** the length of the third side.

2. $\triangle FBK$ with midsegment $\overline{SR}$ and trapezoid $FTNK$ are shown where $\overline{FT} \parallel \overline{KN}$.



- a. Explain why $SF = KS$.
 $\overline{SR}$ is the midsegment of $\triangle FBK$, so point S is the midpoint of $\overline{FK}$. By definition of a midpoint, $SF = KS$.



Students are introduced to the midsegment of a triangle. The triangle midsegment theorem will be explored more in a later unit. During this lesson, students just need a surface understanding that the midsegment exists, and its relationship to the base of the triangle.



Before moving into question 2, lead students in recalling the properties of a trapezoid and what defines a quadrilateral as a trapezoid.

8.7.3 Guided Instruction: Trapezoid Midsegment Theorem (continued)

- b. If $\overline{NB} \cong \overline{FT}$, justify why $\overline{NR} \cong \overline{RT}$.
 $\angle FTR \cong \angle BNR$ because when parallel lines are crossed by a transversal alternate interior angles are congruent. $\angle TRF \cong \angle NRB$ because vertical angles are congruent. $\triangle FTR \cong \triangle BNR$ by AAS congruence. $\overline{NR} \cong \overline{RT}$ because of CPCTC.
- c. Discuss with your partner what information is necessary so that the definition of a trapezoid can be used to state that $FTRS$ is a trapezoid. Summarize your discussion.
At least one pair of sides must be parallel.
- d. Discuss with your partner what else needs to be determined so that it can be stated that $\overline{SR}$ is a midsegment of $FKNK$. Summarize your discussion.
 S is the midpoint of $\overline{KF}$ and R is the midpoint of $\overline{TN}$



Trapezoid Midsegment Theorem

The midsegment of a trapezoid is parallel to the bases and is half the sum of the bases.

8.7.4 Guided Practice: Using the Trapezoid Midsegment Theorem

Component Overview: Students apply Trapezoid Midsegment Theorem to find missing lengths.



8.7.4 Guided Practice: Using the Trapezoid Midsegment Theorem

1. Trapezoid $WDLY$ with midsegment $\overline{FZ}$ is shown.



- a. If $WD = 14.2$ and $YL = 29.7$, what is the length of $\overline{FZ}$?

$$FZ = \frac{1}{2}(WD + YL) = \frac{1}{2}(14.2 + 29.7) = 21.95$$

- b. If $WDLY$ is an isosceles trapezoid and $m\angle DLY = 53^\circ$, what is $m\angle WFZ$?

$$m\angle WFZ = 53^\circ$$

- c. If $FZ = 11.4$ and $YL = 17.6$, what is the length of $\overline{WD}$?

$$FZ = \frac{1}{2}(WD + YL)$$

$$11.4 = \frac{1}{2}(WD + 17.6)$$

$$WD = 5.2$$

- d. What is the value of k if $WD = 4k + 4.2$, $FZ = 7k + 2.1$, and $YL = 6k + 16.8$?

$$FZ = \frac{1}{2}(WD + YL)$$

$$7k + 2.1 = \frac{1}{2}((4k + 4.2) + (6k + 16.8))$$

$$k = 4.2$$

- e. If $WF = \frac{2}{3}p$ and $WY = 12 - 2p$, what is the length of $\overline{WY}$?

$$2(WF) = WY$$

$$2\left(\frac{2}{3}p\right) = 12 - 2p$$

$$p = 3.6$$

$$WY = 12 - 2(3.6) = 4.8$$



Each part of question 1 is independent (question 1 parts a – e can be completed independently). Consider posting the questions around the room for a mini Gallery Walk. Students can be broken into groups to work on a question and then rotate to check the work and make comments on the other questions.



MA.K12.MTR.3.1: Mathematical Fluency

Students need the opportunity to develop their own methods and explore other ways of thinking to build flexibility in fluency.

- Why did you take that approach?
- What would happen if you tried this as your first step?
- Could there be other answers?
- How do you know that (pick a number) is not another possible answer?
- What does this tell you about the trapezoid?

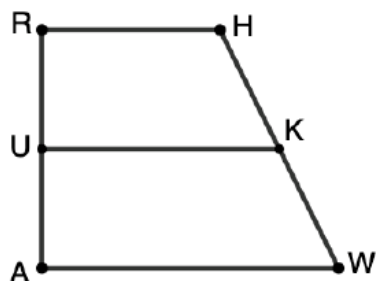
8.7.5 Your Turn!

Component Overview: Students engage in independent practice, applying properties of trapezoids to solve problems.



8.7.5 Your Turn!

1. Trapezoid $RHWA$ is shown with midsegment $\overline{UK}$. Find the values of x and y if $RU = -2x - 2y$, $AU = 10$, $RH = 4x$, $UK = 14$, and $AW = -y + 8$.



$$RU = AU$$

$$-2x - 2y = 10$$

$$UK = \frac{1}{2}(RH + AW)$$

$$14 = \frac{1}{2}(4x + (-y + 8))$$

$$10 = 2x - \frac{1}{2}y$$

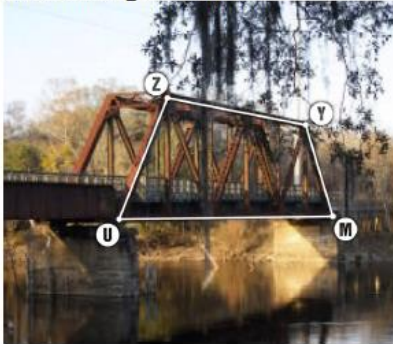
$$x = 3, y = -8$$



Question 1 involves solving a system of equations. Students may need a brief review of how to solve a system of equations in two variables.

8.7.5 Your Turn! (continued)

2. Located on the Nature Coast State Trail in north-central Florida is the Suwannee River Trestle, a bridge that is shaped like an isosceles trapezoid. The state has been inspecting older bridges to ensure their safety and has determined that the Suwannee River Trestle, represented as trapezoid $ZYMU$, needs an additional support brace through its middle.



The support brace is to connect sides $\overline{ZU}$ and $\overline{YM}$ at their midpoints. If $ZU = 28$ feet (ft) and the perimeter of the trestle measures 153 ft, what is the length, in ft, of the new support brace?

$$\begin{aligned}\text{Perimeter} &= ZU + YM + ZY + UM \\ 153 &= 28 + 28 + ZY + UM \\ 97 &= ZY + UM\end{aligned}$$

$$\begin{aligned}\text{Midsegment} &= \frac{1}{2}(ZY + UM) \\ \text{Midsegment} &= \frac{1}{2}(97) \\ \text{Midsegment} &= 48.5 \text{ ft}\end{aligned}$$



Three Reads

Consider prompting students to engage in Three Reads if they are struggling to decode the word problem on their own.



What information in the question 2 stem is needed to solve this problem?

8.7.6 Wrap-Up: Cool Down

Component Overview: Students capture their learning by writing a note to an absent classmate.



8.7.6 Wrap-Up: Cool Down

1. Your friend is absent today and will need the notes from class. Write a quick note to help them understand the trapezoid midsegment theorem.

Answers will vary. The midsegment of a trapezoid is the line segment that connects the midpoints of the two legs of the trapezoid. A midsegment is parallel to the bases of the trapezoid and its length is equal to half the sum of the length of the two bases.



Use the summaries written here to determine understanding and remediation needs for this lesson.